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A machine learning approach in stress testing US bank holding companies☆

Ahmadou Mustapha Fonton Moffo

Université du Québec à Montréal, 315 Sainte-Catherine Street, East, H2X 3X2, Montreal, Canada

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ABSTRACT

This paper assesses the utility of machine learning (ML) techniques combined with comprehensive macroeconomic and microeconomic datasets in enhancing risk analysis during stress tests. The analysis unfolds in two stages. I initially benchmark ML's efficacy in forecasting two pivotal banking variables, net charge-off (NCO) and pre-provision net revenue (PPNR), against traditional linear models. Results underscore the superiority of Random Forest and Adaptive Lasso models in this context. Subsequently, I use these models to project PPNR and NCO for selected bank holding companies under adverse stress scenarios. This exercise feeds into the Tier 1 common equity capital (T1CR) densities simulation. T1CR is the equity capital ratio corrected by some regulatory adjustments to risk-weighted assets. Crucially, findings reveal a pronounced left skew in the T1CR distribution for globally systemically important banks vis-à-vis linear models. By mirroring distress akin to the Great Recession, ML models elucidate intricate macro-financial linkages and enhance risk assessment in downturns.

1. Introduction

In the aftermath of the 2007–2009 financial crisis, the Federal Reserve (FED) introduced an enhanced supervision framework for large banks and complex financial institutions, which became a pivotal element of the Dodd-Frank Wall Street Reform and Consumer Protection Act.[1] As part of this framework, the FED conducts an annual stress test, which evaluates how distressed macroeconomic and financial conditions may impact the regulatory capital of bank holding companies (BHCs). This assessment aims to determine whether BHCs, with total consolidated assets of at least 10 billion USD, have enough capital to absorb losses during challenging situations while still providing loans to households and businesses.

This paper focuses on the top-down approach to stress tests, which comprises two stages. In the first stage, the FED constructs forecasting models for pre-provision net revenue (PPNR)[2] and net charge-offs (NCO),[3] two crucial variables that influence the dynamics of a bank's equity capital. In the second stage, these models are used to simulate PPNR and NCO under stressful economic scenarios over a given stress test horizon. The ultimate goal of this exercise is to compute the Tier 1 common equity capital (T1CR) for each selected BHC during the same period. T1CR, the ratio of common equity Tier 1 capital (CET1)[4] to risk-weighted assets is the final output of the stress test. To assess the capital positions of the selected BHCs, the FED compares T1CR to a threshold defined by the Basel III committee throughout the stress test horizon.

I aim to go beyond the traditional point forecasting of T1CR. Specifically, I seek to investigate whether integrating machine learning techniques (ML) with comprehensive macroeconomic and microeconomic data sets can improve risk analysis in stress tests by offering a more accurate estimation of the T1CR distribution compared to standard linear models. This investigation is of significant policy interest as it examines potential avenues for improving the models used in stress tests, ultimately contributing to more robust risk assessment methodologies. Building robust models that accurately predict PPNR and NCO under hypothetical macroeconomic scenarios is essential to capture the risk in stress tests. The accurate prediction of these variables is crucial for constructing a reliable estimation of the T1CR distribution, allowing

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E-mail address: fonton_moffo.ahmadou_mustapha@courrier.uqam.ca.

[1] The Dodd-Frank Act, a critical regulatory measure for the American financial system since the mid-1930s, was implemented in July 2010 by President Obama to address the crisis and mitigate future upheavals.

[2] PPNR is the sum of a bank's interest and non-interest income, less interest and non-interest expenses scaled by the bank's average quarterly assets.

[3] NCO represents banks' charges for loan and lease losses, adjusted for recoveries and scaled by the bank's average quarterly assets.

[4] CET1 is the equity capital corrected by regulatory adjustments. It is the core capital and the highest quality capital immediately available to absorb unexpected shocks.

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risk analysis. Liu, Moon, and Schorfheide (2020) underscore the importance of the initial step of this process, which involves creating reliable forecasting models for PPNR and NCO using observed macroeconomic and financial variables. These models are the foundation for subsequent analysis and decision-making within the stress test framework. In light of this, I structure the analysis into two main steps.

In the first step, I investigate to what extent ML and big data can improve the accuracy of PPNR and NCO forecasts compared to standard linear models such as the fixed effect linear model, autoregressive model, random walk model, and pooled OLS model. Previous research (see, for example, Goulet Coulombe, Leroux, Stevanovic, and Surprenant (2022), Liu, Li, Ouyang, Liu, and Wu (2023), Su, Cai, and Wu (2023) and Zhang, Shi, Yang, and Zhou (2021)) has emphasized the ability of ML to capture nonlinearities, making it a valuable tool in macroeconomic and financial forecasting. In the same vein, exploring whether ML techniques can outperform traditional forecasting methods when applied to bank-level variables is essential. In addition, selecting the proper set of predictors enables a more accurate assessment of the financial health of banks and the risks they might encounter. For this investigation, I use a panel of 100 large American BHCs observed from the third quarter of 1986 to the fourth quarter of 2019. I construct various out-of-sample forecasting models, encompassing both linear and ML approaches. The set of predictors aligns with those used by the FED in their stress tests. Given its standard usage by the FED in projecting PPNR and NCO components, I choose the fixed effect linear model as the benchmark for forecasting. Following this, I compute each model's relative out-of-sample mean square error (RMSE), a measure obtained by dividing a particular model's out-of-sample mean square error by that of the fixed effect linear model. The comparison of RMSE values allows the identification of the model offering the most accurate forecasts, indicated by the lowest RMSE. This comparative analysis underscores the added value of machine learning methodologies within this context. In addition, I challenge the significance of expanding the macroeconomic database and integrating BHC balance sheet characteristics beyond the variables used in the FED stress scenarios when forecasting PPNR and NCO. To this end, I sequentially construct ML forecasting models for PPNR and NCO, gradually expanding the set of predictors to include a larger macroeconomic database and subsequently combining this extensive macro data with a wide range of microdata. Using the RMSE criteria, I compare the performance of ML models that incorporate this expanded macroeconomic database against models that strictly rely on the predictors defined in the FED stress scenarios. Moreover, I juxtapose ML models that use extensive macroeconomic and microeconomic data against models that rely solely on enlarged macroeconomic information. This multi-faceted comparison provides valuable insights into whether expanding the pool of predictors results in a noticeable relative improvement in the forecast accuracy of ML models compared to the top-performing linear model.

In the second step, I use the best-performing ML forecast models for NCO and PPNR to generate conditional predictions under severely adverse scenarios defined by the FED. From these predictions, I compute point forecasts of T1CR and simulate T1CR distribution for each selected BHC throughout the stress test horizon. Therefore, using ML modeling, I can effectively characterize the banking sector's vulnerability and assess capital adequacy risks. For comparison purposes, I replicate the same analysis using a fixed effect linear model as the forecasting model for NCO and PPNR. In addition, by assessing the density forecast accuracy of both the ML model and the fixed effect linear model during the Great Recession and comparing the distribution of T1CR for globally systematically important banks, I may gain a deeper understanding of their respective abilities to capture and predict risks in a challenging economic environment. This approach provides valuable insights into the potential impact of adverse scenarios on the financial stability of individual BHCs and the banking system as a whole.

The main contribution of this paper is to showcase the value of integrating ML modeling into the risk analysis of stress tests, both indirectly and directly. Indirectly, the paper explores various models and demonstrates that ML models improve the forecast accuracy of crucial banking variables. It also sheds light on the importance of expanding macroeconomic information in improving NCO and PPNR forecast accuracy over some time horizons. This improvement in accuracy can have significant implications for assessing and managing risk in the banking sector. Directly, the paper highlights that ML modeling also leads to an improved density forecast of T1CR under stressful conditions compared to linear models. Many prior studies do not consider ML modeling in predicting NCO and PPNR (see, for example, Guerrieri and Welch (2012), Hirtle, Kovner, Vickery, and Bhanot (2016) and Liu et al. (2020)). Indeed, while some studies have applied machine learning techniques to forecast PPNR and NCO, such as the works by Barth, Emrich, and Güllich (2019) and Kapinos and Mitnik (2016), they do not consider the broader implications for risk analysis and financial stability.

The rest of the paper is organized as follows. Section 2 provides a brief overview of the FED stress test framework. Section 3 presents the methodology. Section 4 contains the description of the data. Section 5 delivers results. Section 6 discusses the policy implications of the results, and Section 7 concludes.

2. An overview of the FED stress test framework

Before the financial crisis, stress tests were typically discretionary and limited to specific institutions.[5] In the wake of the Great Recession, monetary authorities in developed and emerging countries established rules that made stress tests mandatory for large banks and complex financial institutions.[6] The primary goal of these stress tests is to mitigate the impact of potential economic downturns on the financial system and the real economy. Since 2011, the FED in the USA has been conducting annual stress tests on large banks and financial institutions, following a provision of the Dodd-Frank Wall Street Reform and Consumer Protection Act. The main objective of this exercise is to ensure that these institutions maintain sufficient capital to withstand severely adverse economic conditions without disrupting the economy's financing. The FED uses two complementary tools for its stress tests: the Dodd-Frank Act Stress Test (DFAST) and the Comprehensive Capital Analysis and Review (CCAR). These tools assess the capital position of BHCs with assets exceeding ten billion USD. They project critical revenue and loss variables under three scenarios: severely adverse, adverse, and baseline. The severely adverse and adverse scenarios represent hypothetical distressed macroeconomic and financial conditions designed to measure the solidity and resilience of large BHCs. On the other hand, the baseline scenarios rely on unconditional forecasts of scenario variables over a specific time horizon. DFAST and CCAR are closely aligned, employing identical scenario variables, models, and methodologies to produce comparable outcomes.

However, the computation of regulatory capital differs between the two approaches. DFAST uses planned capital distributions, such as dividend payments and common stock repurchases, as defined by the FED, while CCAR uses the bank's planned capital distribution. To pass a stress test, a participating bank should have its projected tier 1 common equity capital ratio (T1CR)[7] remaining greater than or equal

[5] See Lederman (1990) and Hirtle and Lehnert (2015), for examples, of the sectorial stress test in the USA before the Great Recession.

[6] Since 2010, the European Bank Authority has conducted yearly stress tests for European Union member states. The FED made these tests mandatory in the USA and oversaw the process. Other emerging countries, such as Brazil, also conduct regular stress tests.

[7] T1CR is the ratio of common equity tier 1 capital to risk-weighted assets. Mapping book equity capital into common equity tier 1 requires subtracting regulatory adjustments, which include goodwill and other intangibles, deferred tax assets, cash flow from hedge reserves, and gains on sales related to securitization transactions.

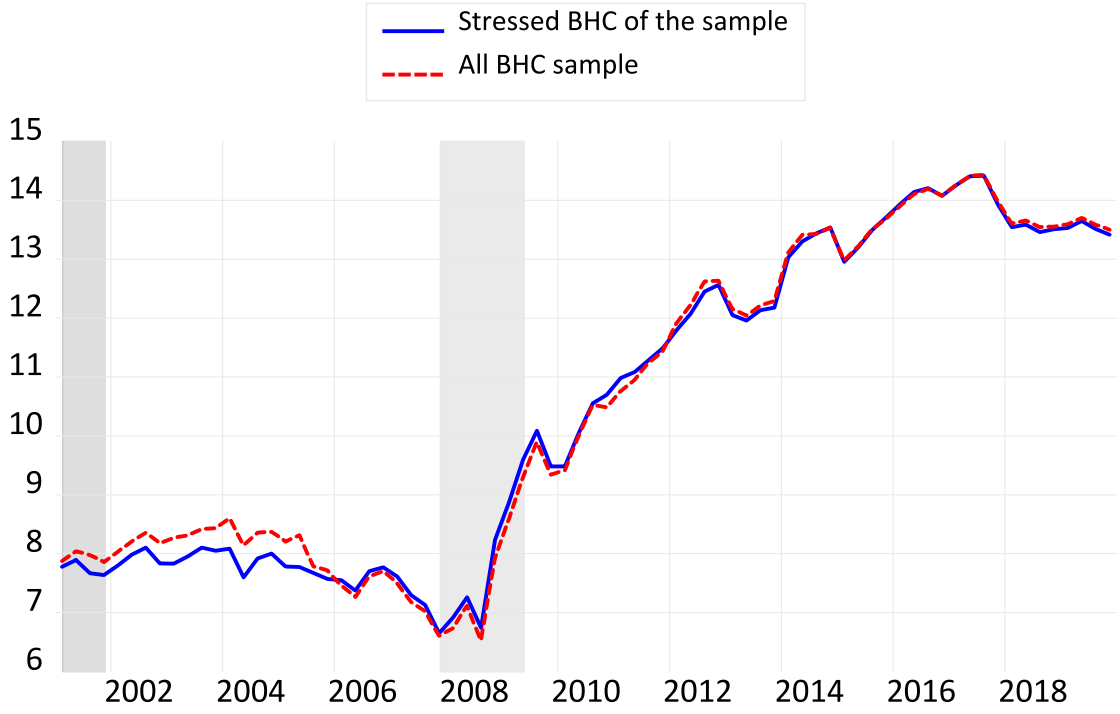


Fig. 1. Aggregate T1CR. Note: The hatched bands represent recessions.

to 4.5% throughout the nine-quarter forecast horizon. Fig. 1 illustrates the evolution of the aggregate T1CR over time, defined as the aggregate common equity tier 1 capital to the aggregate risk-weighted asset. The graph indicates that during the Great Recession, the aggregate T1CR declined, reaching a minimum level close to the regulatory requirement of 4.5%. However, there was a reversal in the trend after the turmoil, with the T1CR steeply increasing and showing moderate stability in 2018. This observation suggests a continuous improvement in banks' capital position since the crisis, with the 4.5% T1CR constraint remaining far from binding. Apart from the T1CR, Basel III sets the minimum total regulatory capital ratio[8] at 8%.

3. Econometric approach

This section describes how I build the dynamic forecast models of two banking measures, i.e., PPNR and the NCO, and use them in the stress test.

3.1. Forecasting methodology

3.1.1. Machine learning techniques

I apply the machine learning techniques to the following models:

$$Y_{i,t+h} = \hat{g}(Z_{i,t}) + e_{i,t+h}\,, \tag{1}$$

where h represents the forecast horizon ranging from one to nine quarters, in line with the range used by the FED in stress test exercises on BHCs. $Y_{i,t+h}$ denotes the value of either PPNR or NCO of BHC i at quarter $t+h$, $e_{i,t+h}$ represents the forecast error and $Z_{i,t}$ is a $K \times 1$ regressor vector. The h-step ahead forecast of variable $Y_{i,t+h}$ for BHC i given the information available at t is given by $\hat{Y}_{i,t+h} = \hat{g}(Z_{i,t})$. I opt for a direct forecast approach for h = 2 to 9, as it has been shown in various studies that it is more robust to misspecification compared to iterated forecasts (Chevillon & Hendry, 2005; Marcellino, Stock, & Watson, 2006; Schorfheide, 2005). Given a lack of confidence in a particular model, I explore several models to find the most suitable. The forecast is dynamic as I use the predictions of the lagged dependent variable among explanatory variables for horizons h = 2 to 9. I train $\hat{g}$ using three specifications combining parametric and nonparametric machine learning techniques. The parametric techniques I consider include "Lasso" regression, "Adaptive Lasso" regression, "Ridge" regression, "Elastic Net" regression, and Principal Component Analysis (PCA). I include Gradient Boosting, Random Forest, and Neural Networks for nonparametric techniques.

In the first specification, $Z_{i,t}$ includes lagged dependent variables, some bank-specific predictors, and the macroeconomic and financial series used by the FED in their stress tests. The second specification expands $Z_{i,t}$ to include more macroeconomic and financial series. The last specification expands the second specification's $Z_{i,t}$ to incorporate more bank-specific predictors. The primary objective of all machine learning techniques is to train $\hat{g}$, which minimizes the out-of-sample Mean Squared Error (MSE) while simultaneously applying regularization procedures. Regularization serves a dual purpose. Firstly, it prevents the overfitting of the data, where the estimated function $\hat{g}$ fits the variable of interest too closely, leading to poor out-of-sample prediction performance. Secondly, it reduces the model complexity and often results in better forecast performance than traditional techniques through bias–variance trade-offs. The choice of appropriate hyperparameters is crucial in regularization procedures. These hyperparameters include penalization coefficients in "Lasso", "Elastic Net", and "Ridge" regressions, as well as the number of trees and variables selected randomly at each step for Random Forest.

I determine the optimal hyperparameters through K-fold cross-validation. Since the sample is a panel dataset, I perform cross-validation across time series and cross-sectional dimensions. Specifically, I randomly divide the subsample into k groups of approximately equal size for a given model and a combination of hyperparameters. Group j where $(1 \leqslant j \leqslant k)$ is treated as the validation set, and I estimate the model on the remaining $k-1$ groups. Then, I predict the variable of interest PPNR or NCO for group j. The Mean Squared Error (MSE^j)

[8] The total regulatory capital ratio is the sum of core equity tier 1 and tier 2 capital to the risk-weighted asset. Tier 2 capital is the second layer of capital, comprising revaluation reserves, hybrid instruments, and subordinated term debt.

is computed as:

$$MSE^j = \frac{1}{N}\sum_{i,t}(Y_{i,t+1} - \hat{Y}^j_{i,t+1})^2,$$

where N represents the number of banks in the validation set, i denotes the cross-sectional dimension, and t represents the time dimension. This process is repeated k times, treating each of the remaining $k-1$ groups as a validation sample. Finally, I compute the average forecasting performance for a given set of hyperparameters as follows:

$$MSE = \frac{1}{N}\sum_{1}^{k} MSE^l,$$

where l ranges from 1 to k. I repeat this process for all potential combinations of hyperparameters and select the ones that yield the lowest MSE, thus deriving the training model. I perform a grid search to explore different combinations of hyperparameters to derive optimal ones. The main limit of this method is the computer cost. However, this limit is overcome with the high performance of computers. Moreover, the likelihood of missing the optimal hyperparameter is low.

3.1.2. Estimation of standard linear forecasting models

In addition to machine learning models, I consider several standard linear forecast models: the autoregressive model, the random walk model without drift, the pooled OLS model, and the fixed effect linear model. They can be defined as follows:

$$Y_{i,t+h} = \alpha + \rho Y_{i,t} + \epsilon_{i,t+h}\,, \tag{2}$$

$$Y_{i,t+h} = Y_{i,t} + \epsilon_{i,t+h}\,, \tag{3}$$

$$Y_{i,t+h} = \alpha + \rho Y_{i,t} + Z'_{i,t}\beta + \epsilon_{i,t+h}, \tag{4}$$

$$Y_{i,t+h} = \alpha_i + \rho Y_{i,t} + Z'_{i,t}\beta + \epsilon_{i,t+h}. \tag{5}$$

In these equations, $Z_{i,t}$ represents the regressor vector defined in Eq. (1) but relative to first specification. Parameters α, ρ, and β are estimated using OLS, while α_i represents the bank fixed effect. Eq. (5) is estimated using the Least Squares Dummy Variable (LSDV) estimator. It is worth noting that the estimation of Eq. (5) can be biased in the presence of lagged dependent variables, as discussed in Galvao (2011). However, this bias diminishes in a long panel dataset like ours. Although I define the fixed effect linear model as the benchmark, the other linear models also serve as natural benchmarks. These linear models are widely used in the literature on bank stress tests, as highlighted in studies such as Covas, Rump, and Zakrajšek (2014), Hirtle et al. (2016), and Liu et al. (2020). By comparing the machine learning models to these linear models, I can assess the performance and contribution of ML models in the context of bank stress tests.

3.1.3. Pseudo out-of-sample forecasting exercise

I use the expanding window estimation technique for the dynamic forecasting exercise. Initially, I divide the data sample into two parts, i.e., the estimation and validation sample, which spans from 1986q3 to 2000q1, and the test set, which covers the period from 2000q2 to 2019q4. After selecting the hyperparameters, I estimate the model using the entire first subsample for each machine learning (ML) method concerning all three specifications. The goal is to forecast the dependent variable $Y_{i,t+h}$, h quarters ahead. I iteratively run the process by expanding the estimation and validation sample by one quarter until I have predicted the dependent variable for the final quarter of the test set, 2019q4. I follow a similar procedure to build linear standard forecast models (SL) but estimate them based only on the first specification. Once I construct various forecast models, I need to evaluate their forecast accuracy.

Let M denotes the number of competing forecast models, h represents the forecast horizon ($h = 1...9$), and j indicates the specification ranging from 1 to 3. To evaluate the forecast model K ($K = 1...M$) at the forecast horizon h within specification j, I calculate its relative out-of-sample mean square error ($RMSE^K_j(h)$), which is defined as :

$$RMSE^K_j(h) = \frac{\sum_{i=1}^{N}\sum_{t=T_{min}}^{T_{max}}(Y_{i,t+h} - \hat{Y}^K_{i,t+h}(j))^2}{\sum_{i=1}^{N}\sum_{t=T_{min}}^{T_{max}}(Y_{i,t+h} - \hat{Y}^B_{i,t+h})^2}\,.$$

Here, N, T_{min}, and T_{max} represent the number of banks, the start date, and the end date of the test set sample, respectively. $Y_{i,t+h}$ denotes the value of the variable of interest for bank i at quarter $t+h$ while $\hat{Y}^K_{i,t+h}(j)$ represent predicted value of $Y_{i,t+h}$ h quarters ahead relative to model K and given information at quarter t. $\hat{Y}^B_{i,t+h}$ represents the predicted value of $Y_{i,t+h}$ h quarters ahead relative to the benchmark model and given information at quarter t. The numerator represents the out-of-sample mean square error ($MSE^K_j(h)$) for model K at horizon h within specification j. At the same time, the denominator refers to the same quantity as the numerator but relative to the benchmark model.

I identify the best model within each specification and for a given forecast horizon by following a two-step process.[9] First, I determine the best ML and SL model.[10] Second, I compare the best SL model with the best ML model. The model with the lowest RMSE is the superior model. I use the Diebold–Mariano test, adapted for panel data, to rank them formally. Diebold and Mariano (2002) provide the test (the Diebold Mariano test) adapted for panel data, allowing the models to be ranked. In my case, I use this test to compare the forecast errors of the best SL and ML models.[11] Besides determining the best forecast model, this comparison exercise helps assess the relevance of ML techniques in forecasting banking variables by comparing the best ML and SL models under specification 1. Indeed, given that both models use the same limited information set, any difference in forecasting performance can be attributed to the forecasting technique itself. Last, I compare ML models across different specifications to assess the contribution of expanding macro and microdata. Specifically, I compare out-of-sample RMSE of ML models across specifications.

3.2. Stress test methodology

In light of the role played by banks in the recent Financial Crisis, many papers focused on the topics on bank distress, bank failure, bank factor crisis, banks and systemic risk using sometimes ML techniques (see, for example, Cleary and Hebb (2016), Danielsson, James, Valenzuela, and Zer (2016a, 2016b), Gogas, Papadimitriou, and Agrapetidou (2018) and Iturriaga and Sanz (2015)). Conversely to these papers, our methodology is forward-looking as I examine how stressed economic and financial scenarios could impact bank solidity and bank contribution to systemic risk. This section describes the methodological approach of the stress test. First, it outlines how I map the predictions of PPNR and NCO under severely adverse scenarios to the prediction of T1CR. Then, I present a formal framework of the stress test exercise.

3.2.1. Mapping PPNR and NCO predictions into the T1CR

The stress test is a conditional forecast exercise. Therefore, I use the best PPNR and NCO forecasting models to predict the two banking variables under the severely adverse scenarios defined by the FED. PPNR and provision on loan losses affect the regulatory capital through

[9] The best model given the forecast horizon has the lowest RMSE. Here, I proceed in two steps to gauge the eventual superiority of ML models over SL models.

[10] An ML model K is the best ML forecast model for specification j relative to a forecast horizon h if for any ML model L estimated in the specification and different from M $RMSE^K_j(h) \leqslant RMSE^L_j(h)$. Similarly, SL model N is the best SL forecasting model if for any other SL model P different from N, $RMSE^N_j(h) \leqslant RMSE^P_j(h)$.

[11] Detailed information about the Diebold–Mariano test can be found in Appendix B.

their direct action on book equity capital. Since, practically, banks adjust their provisions to respond to the variation in charge-off on loan and lease losses, I assume, as Covas et al. (2014), that provision is equal to net charge-off. Moreover, I keep Covas et al. (2014) assumption on the evolution of the book equity capital amount that can be expressed as follow:

$$K_{i,t} = K_{i,t-1} + (1-\tau) \times [PPNR_{i,t} \times \overline{Asset_i} - NCO_{i,t} \times \overline{Asset_i}] - \overline{Equity\ payout_i}\ . \tag{6}$$

$K_{i,t}$ is the dollar amount of book equity capital of BHC i at quarter t. τ is the marginal tax rate set at 21%, its value in 2018q4, the quarter before the stress test horizon 2019q1–2020q1. $PPNR_{i,t}$ and $NCO_{i,t}$ are, respectively, the PPNR and NCO of BHC i at quarter t. The principle behind the stress test is to ensure that large BHCs maintain their capacity to intermediate even when faced with adverse shocks. To translate this principle, I follow the approach of Covas et al. (2014) and assume that the total assets of BHC i at quarter t denoted as $(Asset_{i,t})$, remain constant throughout the stress horizon and equal to its value in 2018q4 $(\overline{Asset_i})$. As Covas et al. (2014), I also assume that the aggregate equity payout of BHC i, which includes dividends paid on common and preferred stocks and the repurchase of Treasury shares, remains constant. I set its value to its corresponding value in 2018q4. By assuming a constant equity payout, I consider the costliness for a bank to raise new equity during a severe recession. The prediction of the common equity tier 1 capital, which is the main focus of the stress test, is determined by the aforementioned predicted book equity capital, the predicted regulatory adjustments denoted as $RADJUST_{i,t}$, and the predicted risk-weighted assets denoted as $RWA_{i,t}$. For the same reasons mentioned earlier, I assume that $RADJUST_{i,t}$ and $RWA_{i,t}$ are constants throughout the stress test horizon and are respectively equal to their values in 2018q4 denoted by $\overline{RADJUST_i}$ and $\overline{RWA_i}$. Formally, the common equity tier 1 capital (T1CR) can be calculated as follows :

$$T1CR_{i,t} = \frac{K_{i,t} - \overline{RADJUST_i}}{\overline{RWA_i}}.$$

3.2.2. Density forecast of T1CR

I study the entire distribution of T1CR instead of focusing only on its mean as usual in the literature. Predicting the distribution of T1CR helps to compute the probability of violating the minimum requirements, one central indicator in stress tests.[12] Therefore, this approach provides a more detailed picture of the banks' vulnerability and risk in an unfavorable macroeconomic environment than the one relying solely on point estimates. Like Covas et al. (2014), I use a two-step approach to forecast the density of T1CR under distressed economic conditions.

In the first step, I use Athey, Tibshirani, and Wager (2019) Quantile Random Forest method to infer conditional quantiles. I adopt a recursive window-expanding estimation technique commonly used in forecasting exercises to perform this estimation. The Quantile Random Forest approach is a nonparametric method that mitigates the issues of misspecification bias. It leverages the splitting tree rules of Random Forest and adapts them to capture quantile heterogeneity effectively. The conditional quantile is chosen to minimize the weighted absolute values of errors as stated below:

$$\hat{q}_\tau(y_{it+1}/x_{i,t}) = argmin_{\beta(x_{i,t})} \left\{ \sum_{i=1}^{N}\sum_{t=1}^{T} \left[\tau(y_{i,t+1} - \beta(x_{i,t}))1(y_{i,t+1} \geqslant \beta(x_{i,t})) + (1-\tau)(\beta(x_{i,t}) - y_{i,t+1})1(y_{i,t+1} < \beta(x_{i,t})) \right] \right\}.$$

N and T denote the data's cross-sectional and time series dimensions. $y_{i,t}$ is the variable of interest, and $x_{i,t}$ is the vector of the forcing variables, the same that the FED uses in its severely adverse scenario. 1(.) denotes an indicator function, τ is the percentile, and $\hat{q}_\tau(y_{it+1}/x_{i,t})$ represents the τ percent quantile. I forecast the 1, 5, 50, and 95 percent conditional quantiles through the stress horizon.

In the second step, to recover the density function, I fit a skewed t-distribution of Azzalini and Capitanio (2003) with the constrain of matching it to the estimated quantile function:

$$f(y;\mu;\sigma;\alpha;\nu) = \frac{2}{\sigma} t\left(\frac{y-\mu}{\sigma};\sigma\right) T\left(\alpha \frac{y-\mu}{\sigma} \sqrt{\frac{\nu+1}{\nu + (\frac{y-\mu}{\sigma})^2}}; \nu+1\right). \tag{7}$$

t(.) denotes the PDF of the Student-t distribution, and T(.) is its corresponding CDF. The four parameters μ, σ, α, and ν pin down the distribution's location, scale, shape, and fatness, respectively. The intuition is to perturb a standard symmetric t-distribution PDF by its CDF, adding an asymmetry through the slant parameter α. For example, if $\alpha = 0$ and $\nu < \infty$, f reduces to standard symmetric t-distribution. The case in which $\alpha = 0$ and $\nu = \infty$ results in a standard normal distribution, while the case in which $\alpha \neq 0$ and $\nu = \infty$ leads to a skewed normal distribution. $\alpha \neq 0$ and $\nu < \infty$ corresponds to a skewed t distribution. This class of distributions offers greater flexibility compared to the standard t-distribution, which has only two parameters. It allows for different shapes and can accommodate both symmetric and asymmetric environments. Due to this flexibility, this family of density functions has been extensively used in the growth and risk literature to examine the asymmetric effects of financial conditions on growth rates. Notable works such as Adrian, Boyarchenko, and Giannone (2019) and Plagborg-Møller, Reichlin, Ricco, and Hasenzagl (2020) have considered this class of functions. In the banking stress test, I use this class of functions to account for potential asymmetry in the distribution of T1CR under disastrous macroeconomic conditions. Covas et al. (2014) previously highlighted the heavy left tail of the density forecast for aggregate T1CR at the end of the stress period, indicating evidence against the symmetry of the T1CR distribution during turmoil.

To fit the t-skewed distribution, I adopt the framework developed by Adrian et al. (2019). For each quarter of the stress horizon and for each stressed bank, I estimate the four parameters $\mu_{i,t+1}$, $\sigma_{i,t+1}$, $\alpha_{i,t+1}$, and $\nu_{i,t+1}$. The estimation is performed by minimizing the squared distance between the estimated conditional quantiles and the quantiles of the t-distribution. Formally, I can state the following optimization problem :

$$\{\mu_{i,t+1}, \sigma_{i,t+1}, \alpha_{i,t+1}, \nu_{i,t+1}\} = argmin_{\mu,\sigma,\alpha,\nu} \sum_{\tau} \left[\hat{q}_\tau(y_{it+1}/x_{i,t}) - F^{-1}(\tau;\mu,\sigma,\alpha,\nu) \right]^2; \tag{8}$$

where $\tau = 1, 5, 50, 95$ are the percentiles, and F is the CDF of the t-skewed distribution defined in (7).

4. Data

I use a panel of 100 BHCs, each with total assets consistently exceeding three billion USD, to implement the methodology. The panel data spans from 1986q3 to 2019q4, covering significant events such as the implementation of Basel I, II, and III regulations, the 1990 recession, and the recent subprime crisis (see Appendix C for BHCs list). As a result of mergers, acquisitions, and bank failures, the panel is unbalanced-only 20% of BHCs operate throughout the period. In line with the Federal Reserve's stress-testing practices, I focus on BHCs instead of commercial banks. The banks that merges, or being acquired by other entities or fails in bankruptcy cease to exist and is removed from the rest of the sample. Unlike the FED, which only considers banks with total assets exceeding 10 billion USD, I include a larger number of banks in my sample. Extending sample aims to enhance forecast accuracy by constructing a more extensive dataset with broader temporal and cross-sectional dimensions.

[12] See, for example, Covas et al. (2014).

Table 1
List of FED stress test scenarios macroeconomic variables.

FRED mnemonic	Variable name	Scale	Variable definition
GPC1	GDP	Percentage change	US real GDP growth
UNRATE	UNRATE	Level	US unemployment rate
CPIAUCSL	Inflation rate	Percentage change	U.S CPI inflation
TB3MS	TB3MS	Level	U.S. 3-month Treasury rate
GS5	GS5	Level	U.S. 5-year Treasury yield
GS10	GS10	Level	U.S. 10-year Treasury yield
MORTGAGE30US	Mortgagerate	Level	U.S. mortgage rate
SP 500	SP500	Percentage change	U.S. stock market index growth
USSTHPI	USSTHPI	Percentage change	U.S. House Price Index growth(all transactions)
VXOCLSx	VXOCLSx	Level	VIX implied volatility index

Fig. 2 illustrates key data insights. The top left panel shows strong co-movement between the total assets of all BHCs and the sampled BHCs, indicating an accurate representation of the broader BHC population in the sample. Moreover, the top right panel reveals that the sampled BHCs' total assets constitute over 55% of all BHCs' total assets, peaking at 70%. These figures underscore the representativeness of the sample. However, it is important to note a contraction in the total assets of the sampled BHCs relative to all BHCs starting in 2006. This reduction can be ascribed to a regulatory change requiring BHC subsidiaries with total assets over 1 billion USD to become autonomous. This change led to the disappearance of several BHCs from our sample, thereby reducing the total assets of the sampled BHCs.

In the analysis, I focus on two key variables: (1) Net Charge-Off (NCO), defined as the annualized charge-off for loan and lease losses, minus recoveries for these losses, scaled by a quarterly average of total assets; and (2) Pre-Provision Net Revenue (PPNR), defined as the annualized sum of net interest and non-interest income divided by the quarterly average of total assets. These computations draw upon data from the FR Y-9C form of the Consolidated Financial Statements for BHCs, sourced from the Chicago Federal Reserve. Cyclical trends for average PPNR and NCO are evident in the bottom panel of Fig. 2. Average PPNR decreases during recessions while NCO increases, and vice versa, during recoveries. These inverse trends, significant within the context of stress test literature, suggest that stressful macroeconomic conditions can erode a bank's capital position, potentially impeding the effective functioning of the BHC. The goal of stress testing is to assess the impact of such weakened capital positions.

I use two types of predictors: bank-specific and macroeconomic variables across three different specifications to forecast PPNR and NCO accurately. The bank-specific predictors are balance sheet variables drawn from the FR Y-9C form. The macroeconomic predictors, consisting of macroeconomic and financial variables, are sourced from the FRED QD database. These data are transformed into stationary series following the procedures specified in the FRED database. I use the entire series from the FRED QD database as macroeconomic predictors in the second and third specifications, aiming to capture a broad spectrum of potential influences on the variables of interest. However, for the first specification, I intentionally limit the macroeconomic predictors to those defined as FED stress scenario variables (as outlined in Table 1) to precisely assess the effect of these stress indicators and ML modeling in improving PPNR and NCO forecast accuracy. Additionally, to ensure that the sample closely mirrors the banks likely to be of interest in real-world stress tests, I refine it to include only the 13 BHCs that were part of the 2017, 2018, and 2019 stress tests (as listed in Table 2). These BHCs represent 92% of the total assets of all BHCs in the sample, emphasizing large banks' relevance in stress testing scenarios.

5. Results

This section discusses the results of the empirical analysis. I start with some forecasting exercises based on observed data, then check the results' robustness and determine the main drivers of PPNR and NCO. The last subsection is devoted to the empirical stress test.

Table 2
Subsample of BHC included in 2017, 2018 and 2019 stress test.

Bank holding companies	Total Asset in 2018q4[a]	Weight in 2018q4[b]
Bank of America Corporation	2.36e+09	21.40
Bancwest corporation	20 695 678	0.19
BBVA USA Bancshares Inc	90 947 174	0.83
Citigroup INC	1.92e+09	17.41
Comerica Inc	70 906 003	0.64
Fifth Third Bancorp	1.46e+08	1.33
JP Morgan chase and company	2.62e+09	23.82
Keycorp	1.40e+08	1.27
Huntington Bancshares Inc	1.09e+08	0.99
Northern trust corporation	1.32e+08	1.20
PNC Financial Services INC	3.82e+08	3.47
State street corporation	2.45e+08	2.22
Wells Fargo et Company	1.89e+09	17.22
All	1.03e+10	91.99

Note: Total consolidated assets (in thousand dollars). Weight represents the ratio of bank's total asset to all sampled BHC total asset in 2018q4.

5.1. Forecasting exercise

5.1.1. Out-of-sample performance of NCO's predictions models

Tables 3, 4, and 5 present the out-of-sample relative mean square error (RMSE) of NCO forecasting models in specifications (1), (2), and (3). As Table 3 suggests, most machine learning (ML) models outperform the benchmark model in NCO forecasting, as indicated by RMSE values lower than one across various horizons. Specifically, the Diebold–Mariano test (DM) highlights that the Random Forest model significantly surpasses the pooled OLS - the most effective standard linear (SL) model - across the nine forecast horizons in the specification (1) at a 1% significance level. These findings indicate that the Random Forest model offers superior forecasting across all horizons in the specification (1). This superior performance suggests the existence of nonlinearity in the bank's loss model, which linear models may fail to capture adequately. The implication is that ML models like Random Forest, capable of handling complex, nonlinear relationships, may yield more accurate and robust forecasting in economic scenarios like stress testing.

Tables 4 and 6 illustrate how incorporating a broader macroeconomic database instead of limited macro data can impact model rankings and predictive accuracy. As Table 4 indicates, Random Forest outperforms most SL models across most horizons, except for horizons 3 and 8. Despite Gradient Boosting emerging as the best model in Horizon 4, the Diebold–Mariano (DM) test does not establish its superiority over Random Forest. Table 6 reveals that excluding horizons 3 and 8, Random Forest paired with a comprehensive macroeconomic database improves the RMSE of Pooled OLS by an average of 49%. This finding contrasts with an average improvement of 39% when Random Forest uses limited macroeconomic data. These results emphasize the advantages of extending macroeconomic data beyond the Federal Reserve's stress test scenario: it bolsters the forecasting performance of leading ML models in predicting NCO by approximately 10% across seven of the nine forecast horizons. Interestingly, Tables 5 and 6 suggest that the

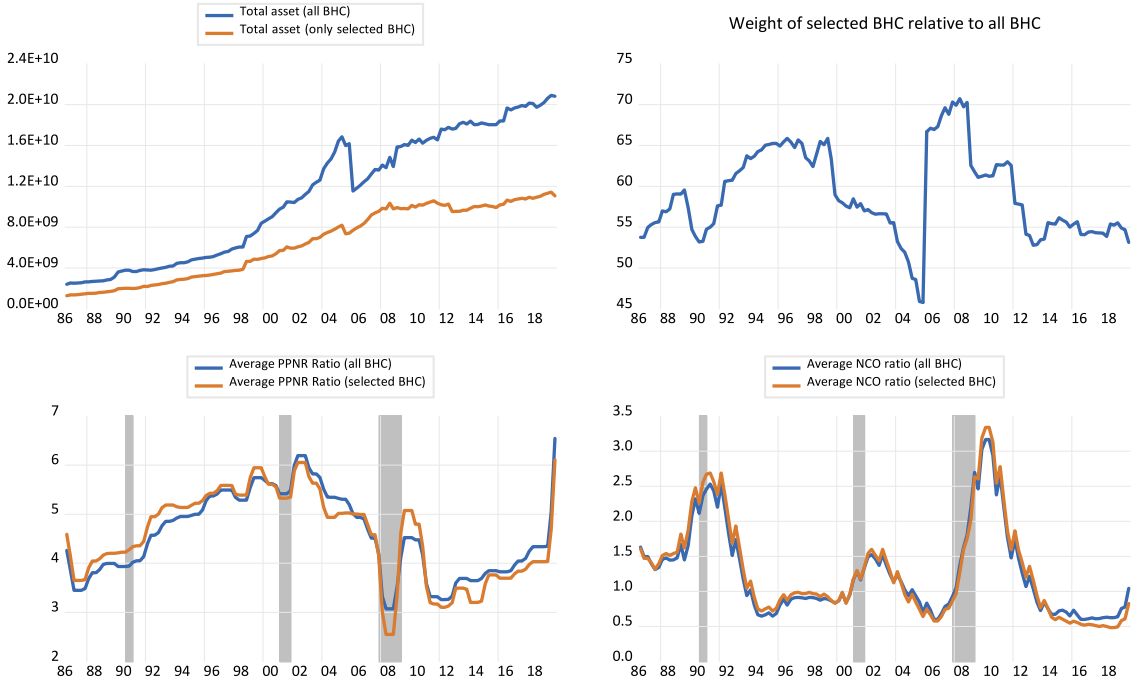


Fig. 2. A brief overview of some data. Note: (1) The hatched bands represent recessions. (2) The average PPNR and NCO are computed as the aggregate PPNR and NCO ratio to total quarterly average assets.

Table 3
Out-of-sample RMSE of NCO's forecast models in specification (1) across various forecast horizons and different forecasting techniques (all banks).

Model	h = 1	h = 2	h = 3	h = 4	h = 5	h = 6	h = 7	h = 8	h = 9
Benchmark model	*1*	*1*	*1*	*1*	*1*	*1*	*1*	*1*	*1*
Pooled OLS	0.62	0.68	0.64	0.71	0.59	0.66	0.66	0.71	0.60
Autoregressive model	2.05	1.93	2.08	3.20	2.05	1.04	1.17	3.08	1.27
Random walk	2.39	1.64	1.61	3.19	1.61	1.03	1.03	1.53	1.21
Lasso	1.51	0.69	0.64	0.68	0.57	0.64	0.65	0.69	0.60
Adaptive Lasso	0.61	0.60	0.65	0.68	0.58	0.66	0.67	0.68	0.62
Ridge	0.59	0.62	0.53	0.97	0.65	0.59	0.53	0.70	0.68
Elastic net	0.64	0.61	0.63	0.68	0.57	0.67	0.65	0.69	0.60
Principal component	2.50	1.71	1.63	3.23	0.58	1.17	1.21	0.62	0.58
NN1	3.39	1.09	0.72	1.58	0.72	0.72	0.72	1.26	0.78
NN2	6.45	1.11	0.67	1.16	0.64	0.70	0.64	0.93	0.76
NN3	4.14	1.17	0.69	1.29	0.65	0.80	0.61	0.91	0.81
Gradient boosting	0.97	0.93	0.71	0.52	0.55	0.43	0.38	0.67	0.45
Random Forest	**0.39*****	**0.45*****	**0.51*****	**0.45*****	**0.35*****	**0.31*****	**0.36*****	**0.36*****	**0.39*****

Note: Machine learning and standard linear models include the same variables FED uses in the bank's stress test. (2) The first line represents the benchmark model's mean square error (MSE) normalized to 1, while the other lines present the relative MSE of different forecast models. (3) The figures in bold relate to the best forecast model. No star: best linear models outperform the best machine learning model or have the same performance as it for the DM test, *: best machine learning model outperforms best standard linear model for the DM test at 10% level, **: 5% level; ***: 1% level. NN1 designates the neural network with one hidden layer that has 32 neurons. NN2 is the neural network with two hidden layers containing 32 and 16 neurons, respectively. NN3 is a neural network with three hidden layers having 32, 16, and 8 neurons, respectively.

inclusion of more bank-specific data does not enhance the predictive accuracy of Random Forest or other ML models. On the contrary, it tends to deteriorate the performance of top ML models relative to the most efficient SL model. This result implies that additional bank-specific data may not offer predictive value beyond what the macroeconomic data and initial bank characteristics already provide.

5.1.2. *Out-of-sample performance of PPNR's predictions models*

Tables 7, 8, 9, and 10 display the out-of-sample RMSE of models across specifications (1), (2), and (3), as well as the percentage variation of the best ML RMSE compared to the best SL RMSE for each specification. Table 7 indicates that the Adaptive Lasso, a linear ML model, demonstrates superior forecasting performance for PPNR across horizons 1 to 8, exhibiting the lowest RMSE. Its forecasting performance surpasses the leading SL models (Pooled OLS or Autoregressive) at a 1% significance level. However, in horizon 9, although the Adaptive Lasso's RMSE remains lower than that of the top-performing SL model, the performance difference is not statistically significant. Other ML models, excluding Principal Component, show potential for superior performance, with RMSE values under one across most forecast horizons. Nevertheless, these models still lag behind the best SL model across all horizons.

Table 8 underscores that by broadening the macroeconomic database beyond the Federal Reserve scenario, three machine learning models stand out across all nine forecast horizons: Adaptive Lasso, Random Forest, and Gradient Boosting. Consistently, Table 10 indicates that including additional macroeconomic information only boosts the accuracy of the top-performing ML model for the more distant horizons (5, 6, and 7). In these instances, the improvement in RMSE of the best ML model over the top SL models is more pronounced than in

Table 4
Out-of-sample RMSE of NCO's forecast models in specification (2) across various forecast horizons and different forecasting techniques (all banks).

Model	h = 1	h = 2	h = 3	h = 4	h = 5	h = 6	h = 7	h = 8	h = 9
Benchmark model	*1*	*1*	*1*	*1*	*1*	*1*	*1*	*1*	*1*
Pooled OLS	0.62	0.68	**0.64**	0.71	0.59	0.66	0.66	0.71	0.60
Autoregressive model	2.05	1.93	2.08	3.20	2.05	1.04	1.17	3.08	1.27
Random walk	2.39	1.64	1.61	3.19	1.61	1.03	1.03	1.53	1.21
Lasso	1.88	3.60	0.97	0.58	1.12	1.59	0.76	0.49	0.67
Adaptive Lasso	2.17	1.28	0.83	0.52	1.17	4.73	0.63	0.54	0.60
Ridge	1.23	2.08	0.80	1.23	1.35	1.50	0.56	0.55	0.66
Elastic net	1.73	3.57	0.78	0.58	1.13	1.69	0.71	0.48	0.66
Principal component	1.13	1.02	0.79	1.39	0.89	0.69	0.67	0.85	0.82
NN1	3.48	1.71	1.74	7.16	1.64	0.87	0.94	1.85	0.98
NN2	3.92	1.96	2.05	8.68	1.91	1.02	1.21	2.38	1.20
NN3	5.08	2.54	2.57	10.55	2.43	1.29	1.38	2.70	1.43
Gradient boosting	0.42	1.05	0.90	**0.39*****	0.28	0.21	0.28	**0.39*****	0.40
Random Forest	**0.28*****	**0.39*****	**0.64**	0.45	**0.22*****	**0.17*****	**0.27*****	0.40	**0.28*****

Note: (1) Machine learning models are estimated with an extensive macroeconomic database.(2) The first line represents the benchmark model's mean square error (MSE) normalized to 1, while the other lines present the relative MSE of different forecast models. (3) The figures in bold relate to the best forecast model. (4) No star: Best linear models outperform the best machine learning model or have the same performance as it for the DM test *: Best machine learning model outperforms best standard linear model for the DM test at 10% level; **: 5% level; ***: 1% level. NN1 designates the neural network with one hidden layer that has 32 neurons. NN2 is the neural network with two hidden layers containing 32 and 16 neurons, respectively. NN3 is a neural network with three hidden layers having 32, 16, and 8 neurons, respectively.

Table 5
Out-of-sample RMSE of NCO's forecast models in specification (3) across various forecast horizons and different forecasting techniques (all banks).

Model	h = 1	h = 2	h = 3	h = 4	h = 5	h = 6	h = 7	h = 8	h = 9
Benchmark model	*1*	*1*	*1*	*1*	*1*	*1*	*1*	*1*	*1*
Pooled OLS	0.62	0.68	**0.64**	**0.71**	**0.59**	0.66	0.66	0.71	**0.60**
Autoregressive model	2.05	1.93	2.08	3.20	2.05	1.04	1.17	3.08	1.27
Random walk	2.39	1.64	1.61	3.19	1.61	1.03	1.03	1.53	1.21
Lasso	1.69	2.83	0.86	0.84	1.16	1.17	0.80	**0.67**	0.80
Adaptive Lasso	2.09	7.99	0.84	**0.71**	1.20	0.94	0.66	**0.67**	0.69
Ridge	1.42	1.05	0.76	1.61	1.38	1.51	0.57	0.76	0.66
Elastic net	1.55	2.79	0.84	0.90	1.17	1.25	0.74	0.70	0.76
Principal component	1.16	1.10	0.98	2.06	1.07	0.79	0.83	1.04	0.87
NN1	5.20	2.57	2.63	10.84	2.50	1.33	1.42	2.78	1.47
NN2	4.05	1.99	2.04	8.45	1.94	1.04	1.12	2.14	1.17
NN3	5.20	2.59	2.63	10.84	2.50	1.33	2.78	0.91	1.49
Gradient boosting	**0.45****	0.61	0.75	1.13	0.68	0.48	0.55	0.84	0.81
Random Forest	0.47	**0.47****	0.69	1.26	0.65	**0.41*****	**0.54*****	0.80	0.62

Note: (1) Machine learning models include an extensive macroeconomic database and enriched BHC characteristics. (2) The first line represents the benchmark model's mean square error (MSE) normalized to 1, while the other lines present the relative MSE of different forecast models.(3) The figures in bold relate to the best forecast model. (4) no star: Best linear models outperform the best machine learning model or have the same performance as it for the DM test *: Best machine learning model outperforms best standard linear model for the DM test at 10% level; **: significance at 5%; ***: significance at 1%. NN1 designates the neural network with one hidden layer that has 32 neurons. NN2 is the neural network with two hidden layers containing 32 and 16 neurons, respectively. NN3 is a neural network with three hidden layers having 32, 16, and 8 neurons, respectively.

Table 6
Percentage variation of best ML relative to best SL model forecast accuracy of NCO under the three specifications (all banks).

Measure	h = 1	h = 2	h = 3	h = 4	h = 5	h = 6	h = 7	h = 8	h = 9
$PC_1(h)$	−37.10	−33.82	−20.31	−36.62	−40.68	−53.03	−45.45	−49.30	−35.00
$PC_2(h)$	−54.84	−42.65	0.00	−45.07	−62.71	−74.24	−59.09	−45.07	−53.33
$PC_3(h)$	−27.42	−30.88	7.81	0.00	10.17	−37.88	−18.18	−5.63	0.00

Note: (1) $PC_j(h)$, j = 1,2,3 denotes the percent variation of best ML relative to best SL forecast accuracy for different forecast horizons h and for specification j. A negative value means that using the best ML reduces the RMSE of the best SL of PC_j, improving its forecast accuracy. A positive value on contrary denotes a deterioration of forecast accuracy in case of using best ML model instead of best SL model.

the first specification. Contrastingly, Tables 9 and 10 reveal that integrating more bank balance sheet data can negatively impact forecast accuracy, leading to an increase in the Adaptive Lasso's RMSE along with other ML models, in comparison to specification (2). Moreover, under specification (3), the top SL model surpasses the best ML model in specific forecast horizons. These results underline the importance of judicious variables and information source selection for each forecast horizon. While adding macroeconomic data can improve the performance of leading ML models in specific horizons, incorporating excessive bank-specific data can prove counterproductive.

5.1.3. Robustness analysis

This subsection evaluates the robustness of the top-performing model selection by focusing solely on the sub-sample of large BHCs selected for the empirical stress test.

Forecasting NCO on the sub-sample of large BHC selected for the empirical stress test

Tables 11 and 12, and 13 delineate the out-of-sample forecasting performance of various models across the three specifications. On the other hand, Table 14 illustrates the percentage variation in RMSE between the top ML model and the most efficient SL model across these specifications. The Random Forest consistently emerges as the

Table 7
Out-of-sample RMSE of PPNR's forecast models under specification (1) across various forecast horizons and different forecasting techniques (all banks).

Model	h = 1	h = 2	h = 3	h = 4	h = 5	h = 6	h = 7	h = 8	h = 9
Benchmark model	*1*	*1*	*1*	*1*	*1*	*1*	*1*	*1*	1
Pooled OLS	0.53	0.50	0.51	0.53	0.51	0.53	0.55	0.63	0.60
Autoregressive model	4.24	1.61	1.31	1.58	0.72	0.53	0.53	0.63	0.40
Random walk	4.53	2.79	2.63	3.28	2.22	1.92	2.00	2.24	1.79
Lasso	0.47	0.45	0.47	0.47	0.47	0.46	0.47	0.54	0.50
Adaptive Lasso	**0.35*****	**0.34*****	**0.39*****	**0.39*****	**0.40*****	**0.40*****	**0.43*****	**0.40*****	**0.38**
Ridge	1.00	0.70	0.67	0.78	0.65	0.61	0.61	0.71	0.66
Elastic net	0.53	0.48	0.51	0.47	0.53	0.52	0.54	0.62	0.60
Principal component	4.47	2.75	2.47	2.67	1.99	1.69	1.76	1.83	1.60
NN1	1.12	0.75	0.71	0.72	0.71	0.73	0.74	0.79	0.87
NN2	1.24	0.80	0.78	0.75	0.72	0.67	0.54	0.90	0.73
NN3	1.47	0.77	0.80	0.78	0.74	0.62	0.71	0.92	0.56
Gradient boosting	0.82	0.41	**0.39**	0.58	0.53	0.80	0.51	0.63	0.49
Random Forest	0.53	0.45	0.47	0.58	0.46	0.43	0.49	0.54	0.44

Note: (1) Machine learning and standard linear models include the same variables FED uses in the bank's stress test. (2) The first line represents the benchmark model's mean square error (MSE) normalized to 1, while the other lines present the relative MSE of different forecast models. (3) The figures in bold relate to the best forecast model. No star: best linear models outperform the best machine learning model or have the same performance as it for the DM test, *: best machine learning model outperforms best standard linear model for the DM test at 10% level, **: 5% level; ***: 1% level. NN1 designates the neural network with one hidden layer that has 32 neurons. NN2 is the neural network with two hidden layers containing 32 and 16 neurons, respectively. NN3 is a neural network with three hidden layers having 32, 16, and 8 neurons, respectively.

Table 8
Out-of-sample RMSE of PPNR's forecast models under specification (2) across various forecast horizons and different forecasting techniques (all banks).

Model	h = 1	h = 2	h = 3	h = 4	h = 5	h = 6	h = 7	h = 8	h = 9
Benchmark model	*1*	*1*	*1*	*1*	*1*	*1*	*1*	*1*	1
Pooled OLS	0.53	0.50	0.51	0.53	0.51	0.53	0.55	0.63	0.60
Autoregressive model	4.24	1.61	1.31	1.58	0.72	0.53	0.53	0.63	0.40
Random walk	4.53	2.79	2.63	3.28	2.22	1.92	2.00	2.24	1.79
Lasso	0.53	0.45	0.76	0.50	0.40	0.41	0.44	0.45	0.41
Adaptive Lasso	**0.41*****	0.39	1.84	**0.39*****	**0.33*****	**0.34*****	0.68	**0.42*****	0.44
Ridge	1.02	0.70	1.00	0.78	0.70	0.63	0.62	0.70	0.65
Elastic net	0.59	0.66	1.41	0.50	0.43	0.52	0.67	0.60	0.42
Principal component	1.24	0.77	0.71	0.78	0.61	0.54	0.59	**0.42**	0.53
NN1	1.53	0.91	0.76	0.83	0.68	0.56	0.58	0.69	0.58
NN2	1.41	0.86	0.83	0.78	0.69	0.54	0.56	0.64	0.56
NN3	1.18	0.95	0.82	0.89	0.71	0.52	0.57	0.71	0.55
Gradient boosting	0.65	**0.34*****	**0.41*****	0.53	0.40	0.48	0.51	0.53	0.44
Random Forest	0.59	0.39	0.43	0.64	0.42	**0.34**	**0.39*****	0.46	**0.35*****

Note: (1) Machine learning models are estimated with an extensive macroeconomic database. (2) The first line represents the benchmark model's mean square error (MSE) normalized to 1, while the other lines present the relative MSE of different forecast models. (3) The figures in bold relate to the best forecast model. (4) No star: Best linear models outperform the best machine learning model or have the same performance as it for the DM test *: Best machine learning model outperforms best standard linear model for the DM test at 10% level; **: 5% level; ***: 1% level. NN1 designates the neural network with one hidden layer that has 32 neurons. NN2 is the neural network with two hidden layers containing 32 and 16 neurons, respectively. NN3 is a neural network with three hidden layers having 32, 16, and 8 neurons, respectively.

best forecasting model across all three specifications, significantly outperforming the Pooled OLS - the leading SL model - in nearly all forecast horizons. Interestingly, Table 14 reveals a more pronounced forecasting performance of the Random Forest relative to the best SL model when forecasting net charge-offs (NCO) for larger banks. For instance, in specification (1), when applied solely to the largest banks, Random Forest improves the RMSE of top SL model by an average of 48.74%, compared to a 39% improvement when applied to the entire sample. Similar patterns are observable in other specifications. One potential explanation could be the relatively poorer performance of the best SL model in forecasting NCO for larger BHCs. Furthermore, the average RMSE of the Random Forest model in specification (2) trails behind the best SL model's RMSE by 57.84% across some horizons. These findings imply that expanding macroeconomic information can enhance the forecasting accuracy of the top ML model for specific forecast horizons. As mentioned previously, when considering the whole sample, broadening the BHC characteristics does not necessarily improve the performance of the best ML model, which underscores the need for a more targeted selection of predictors.

Forecasting PPNR on the sub-sample of large BHC selected for the empirical stress test

As shown by the results from Table 15, the best forecast model of PPNR for the large banks throughout most of the horizons in the specification (1) is the Adaptive Lasso. The usefulness of additional macroeconomic information is limited to very few forecast horizons, in which relative improvement of the ML best model compared to the SL best model is higher than in the first specification (see Tables 16 and 18). Lastly, as revealed in Tables 17 and 18, the supplementary bank-specific data do not lead to forecast amelioration of PPNR relative to the previous specifications.

5.2. Empirical stress test

In this subsection, I use results about forecasting to undertake a pseudo-stress test under severely adverse scenarios. The jump-off date for the conditional forecast is 2018q4, and the stress test period evaluation spans from 2019q1 to 2021q1. The bank's specific forcing variables are assumed to hold constant values equal to their pre-stress levels (2018q4) over the stress horizon. In turn, macroeconomic forcing variables follow the trajectories defined by the FED in its severely adverse scenarios (Table 19).

Table 9
Out-of-sample RMSE of PPNR's forecast models under specification (3) across various forecast horizons and different forecasting techniques (all banks).

Model	h = 1	h = 2	h = 3	h = 4	h = 5	h = 6	h = 7	h = 8	h = 9
Benchmark model	*1*	*1*	*1*	*1*	*1*	*1*	*1*	*1*	1
Pooled OLS	0.53	0.50	**0.51**	0.53	0.51	0.53	0.55	0.63	0.60
Autoregressive model	4.24	1.61	1.31	1.58	0.72	0.53	**0.53**	0.63	**0.40**
Random walk	4.53	2.79	2.63	3.28	2.22	1.92	2.00	2.24	1.79
Lasso	0.53	0.55	0.82	0.50	0.43	0.45	0.66	**0.51*****	0.45
Adaptive Lasso	**0.47*****	**0.45*****	1.88	**0.39*****	**0.39*****	**0.43*****	0.98	0.53	0.53
Ridge	1.03	0.80	1.10	0.78	0.74	0.70	0.70	0.80	0.80
Elastic net	0.59	0.75	1.45	0.50	0.44	0.64	0.78	0.69	0.46
Principal component	1.24	0.89	0.80	0.92	0.67	0.60	0.59	0.63	0.55
NN1	1.53	0.90	0.78	0.83	0.70	0.60	0.71	0.85	0.60
NN2	1.41	0.88	0.80	0.78	0.72	0.58	0.72	0.90	0.58
NN3	1.18	0.97	0.84	0.89	0.74	0.55	0.75	0.95	0.57
Gradient boosting	0.82	0.52	**0.51**	0.67	0.57	0.55	0.62	0.72	0.58
Random Forest	0.65	0.55	0.61	0.92	0.61	0.61	0.64	0.77	0.61

Note: (1) Machine learning models include an extensive macroeconomic database and enriched BHC characteristics. (2) The first line represents the benchmark model's mean square error (MSE) normalized to 1, while the other lines present the relative MSE of different forecast models. (3) The figures in bold relate to the best forecast model. (4) No star: Best linear models outperform the best machine learning model or have the same performance as it for the DM test *: Best machine learning model outperforms best standard linear model for the DM test at 10% level; **: significance at 5%; ***: significance at 1%. NN1 designates the neural network with one hidden layer that has 32 neurons. NN2 is the neural network with two hidden layers containing 32 and 16 neurons, respectively. NN3 is a neural network with three hidden layers having 32, 16, and 8 neurons, respectively.

Table 10
Percentage variation of the best ML relative to the best SL model forecast accuracy of PPNR under the three specifications (all banks).

Measure	h = 1	h = 2	h = 3	h = 4	h = 5	h = 6	h = 7	h = 8	h = 9
$PC_1(h)$	−33.96	−32.00	−23.53	−26.41	−21.57	−24.53	−18.87	−36.51	−5.00
$PC_2(h)$	−22.64	−32.00	−19.61	−26.41	−35.29	−35.85	−26.41	−33.33	−12.50
$PC_3(h)$	−11.32	−10.00	0.00	−26.41	−23.53	−18.87	11.32	−20.63	12.50

Note: $PC_j(h)$, j = 1,2,3 denotes the percent variation of best ML relative to best SL forecast accuracy for different forecast horizons h and for specification j. A negative value means that using the best ML reduces the RMSE of the best SL of PC_j, improving its forecast accuracy. A positive value on contrary denotes a deterioration of forecast accuracy in case of using best ML model.

Table 11
Out-of-sample relative mean square error of NCO's forecast models under specification (1) across various forecast horizons and different forecasting techniques (only banks selected for the empirical stress test).

Model	h = 1	h = 2	h = 3	h = 4	h = 5	h = 6	h = 7	h = 8	h = 9
Benchmark model	*1*	*1*	*1*	*1*	*1*	*1*	*1*	*1*	*1*
Pooled OLS	0.69	0.80	0.75	0.77	0.75	0.79	0.78	0.82	0.72
Autoregressive model	2.02	3.21	3.76	3.40	4.63	1.88	2.41	4.19	3.08
Random walk	2.38	1.81	1.97	1.91	1.68	1.04	0.99	1.58	2.58
Lasso	0.72	0.80	0.73	0.77	0.67	0.75	0.77	0.78	0.72
Adaptive Lasso	0.69	0.79	0.76	0.73	0.67	0.79	0.79	0.77	0.76
Ridge	0.62	0.71	0.55	0.91	0.73	0.66	0.57	0.74	0.82
Elastic net	0.72	0.71	0.72	0.73	0.67	0.76	0.75	0.78	0.72
Principal component	2.70	1.87	1.61	1.64	0.67	1.10	1.07	0.67	0.67
NN1	3.39	1.60	0.86	1.55	0.84	0.94	0.84	1.31	0.93
NN2	6.45	1.54	0.76	1.05	0.72	0.87	0.72	0.89	0.94
NN3	4.14	1.73	0.80	1.18	0.75	1.01	0.70	0.95	1.04
Gradient boosting	1.26	1.54	0.93	0.55	0.84	0.59	0.45	0.94	0.72
Random Forest	**0.25*****	**0.49*****	**0.59*****	**0.30*****	**0.31*****	**0.25*****	**0.30*****	**0.49*****	**0.54*****

Note: (1) Machine learning models and standard linear models include the same variables that FED uses in the bank's stress test. (2) The first line represents the mean square error (MSE) of the benchmark model normalized to 1, while the other lines present the relative MSE of different forecast models. (3) The figures in bold relate to the best forecast model. No star : best linear models outperform the best machine learning model or have the same performance as it for DM test, *: best machine learning model outperforms best standard linear model for the DM test at 10% level , **: 5% level ; ***: 1% level. NN1 designates the neural network with one hidden layer that has 32 neurons. NN2 is the neural network with two hidden layers containing 32 and 16 neurons, respectively. NN3 is a neural network with three hidden layers having 32, 16, and 8 neurons, respectively.

5.2.1. Predicting capital adequacy measure under hypothetical severely adverse scenarios

In this section, I run the stress test using two distinct models. The first model (Best ML), which acts as the baseline, employs the Adaptive Lasso and Random Forest - the top ML forecasting models - to construct conditional predictions of PPNR and NCO. Subsequently, these predictions are used to derive T1CR. The second model (FE-LM) relies on a linear framework - specifically, a fixed-effect linear model - to forecast both banking variables (PPNR and NCO), and, subsequently, T1CR. Table 20 presents the minimum predicted T1CR values throughout the stress test horizon for both models and, for comparison, includes the minimum historical T1CR values during the Great Recession (G.R). The "All" row reflects aggregate T1CR, calculated as the ratio of aggregate common equity capital to aggregate risk-weighted assets.

All the banks in the two frameworks meet the minimum T1CR threshold of 4.5%, thus passing the stress test. However, during the last global financial crisis, two banks - Bancwest Corporation and PNC Financial Services INC - would have fallen below this threshold, indicating a potential failure in the stress test. While the mandatory rule for minimum T1CR requirement was not in effect during the crisis, these results highlight the relatively low capitalization of banks prior to the Great Recession. For example, the average T1CR before the crisis

Table 12
Out-of-sample relative mean square error of NCO's forecast models under specification (2) across various forecast horizons and different forecasting techniques (only banks selected for the empirical stress test).

Model	h = 1	h = 2	h = 3	h = 4	h = 5	h = 6	h = 7	h = 8	h = 9
Benchmark model	*1*	*1*	*1*	*1*	*1*	*1*	*1*	*1*	*1*
Pooled OLS	0.69	0.80	**0.75**	0.77	0.75	0.79	0.78	0.82	0.72
Autoregressive model	2.02	3.21	3.76	3.40	4.63	1.88	2.41	4.19	3.08
Random walk	2.38	1.81	1.97	1.91	1.68	1.04	0.99	1.58	2.58
Lasso	2.77	6.36	1.52	0.59	1.81	2.45	1.02	0.54	1.00
Adaptive Lasso	3.33	2.16	1.25	0.50	1.92	7.70	0.83	0.63	0.88
Ridge	1.00	1.50	0.90	0.70	1.20	2.00	0.70	0.62	0.90
Elastic net	2.60	6.34	1.18	0.59	1.81	2.62	0.92	0.54	0.37
Principal component	1.31	1.33	1.06	1.64	1.04	0.78	0.75	0.74	0.94
NN1	3.50	2.00	1.10	1.00	1.30	1.10	0.70	1.00	0.97
NN2	7.00	1.70	1.00	0.90	1.20	1.20	0.74	0.75	0.99
NN3	5.14	1.90	1.30	0.95	1.00	1.40	0.77	0.77	1.10
Gradient boosting	0.54	1.97	1.62	0.44	0.31	0.25	0.31	0.46	0.53
Random Forest	**0.20*****	**0.54*****	1.13	**0.32** ***	**0.17*****	**0.12*****	**0.30*****	**0.39*****	**0.27*****

Note: (1) Machine learning models are estimated with extensive macroeconomic database.(2) The first line represents the mean square error (MSE) of the benchmark model normalized to 1, while the other lines present the relative MSE of different forecast models. (3) The figures in bold relate to the best forecast model. (4) No star : Best linear models outperform the best machine learning model or have the same performance as it for the DM test *: Best machine learning model outperforms best standard linear model for the DM test at 10% level ; **: 5% level ; ***: 1% level.NN1 designates the neural network with one hidden layer that has 32 neurons. NN2 is the neural network with two hidden layers containing 32 and 16 neurons, respectively. NN3 is a neural network with three hidden layers having 32, 16, and 8 neurons, respectively.

Table 13
Out-of-sample relative mean square error of NCO's forecast models under specification (3) across various forecast horizons and different forecasting techniques (only banks selected for the empirical stress test).

Model	h = 1	h = 2	h = 3	h = 4	h = 5	h = 6	h = 7	h = 8	h = 9
Benchmark model	*1*	*1*	*1*	*1*	*1*	*1*	*1*	*1*	*1*
Pooled OLS	0.69	0.80	0.75	0.77	0.75	0.79	0.78	0.82	0.72
Autoregressive model	2.02	3.21	3.76	3.40	4.63	1.88	2.41	4.19	3.08
Random walk	2.38	1.81	1.97	1.91	1.68	1.04	0.99	1.58	2.58
Lasso	2.51	4.80	1.21	0.82	1.65	1.65	0.92	0.76	1.01
Adaptive Lasso	3.21	3.00	1.24	0.59	1.71	6.70	0.74	0.75	0.63
Ridge	0.98	1.70	0.89	0.69	1.10	1.50	0.60	0.70	0.70
Elastic net	2.36	7.31	1.18	0.86	1.67	1.78	0.79	0.80	0.94
Principal component	1.46	1.23	1.14	2.18	1.11	0.78	0.84	0.95	0.92
NN1	3.00	2.10	1.12	1.30	1.30	1.20	0.60	1.40	0.97
NN2	6.00	1.50	1.02	1.10	1.20	1.10	0.70	0.90	0.99
NN3	4.50	1.75	1.40	1.20	1.10	1.40	0.75	0.80	1.10
Gradient boosting	0.44	0.77	0.94	0.73	0.63	0.41	**0.46**	0.56	0.89
Random Forest	**0.24*****	**0.49*****	**0.70**	**0.44*****	**0.53*****	**0.32*****	0.50	**0.52*****	**0.56*****

Note: (1) Machine learning models include extensive macroeconomic database and enriched BHC characteristics .(2) The first line represents the mean square error (MSE) of the benchmark model normalized to 1, while the other lines present the relative MSE of different forecast models.(3) The figures in bold relate to the best forecast model. (4) no star : Best linear models outperform the best machine learning model or have the same performance as it for the DM test *: Best machine learning model outperforms best standard linear model for the DM test at 10% level; **: significance at 5% ; ***: significance at 1%.NN1 designates the neural network with one hidden layer that has 32 neurons. NN2 is the neural network with two hidden layers containing 32 and 16 neurons, respectively. NN3 is a neural network with three hidden layers having 32, 16, and 8 neurons, respectively.

Table 14
Percentage variation of best ML relative to best SL model forecast accuracy of NCO under the three specifications (only banks selected for the empirical stress test).

Measure	h = 1	h = 2	h = 3	h = 4	h = 5	h = 6	h = 7	h = 8	h = 9
$PC_1(h)$	−63.77	−38.75	−21.33	−61.04	−58.67	−68.35	−61.54	−40.24	−25.00
$PC_2(h)$	−71.01	−32.50	20.00	−58.44	−77.33	−84.81	−61.54	−52.44	−62.50
$PC_3(h)$	−65.22	−38.75	−6.67	−42.86	−29.33	−59.49	−41.02	−36.58	−22.22

Note: $PC_j(h)$, j = 1,2,3 denotes the percent variation of best ML relative to best SL forecast accuracy for different forecast horizons h and for specification j. A negative value means that using the best ML reduces the RMSE of the best SL of PC_j, improving its forecast accuracy. A positive value on contrary denotes a deterioration of forecast accuracy in case of using best ML model instead of best SL model.

was 7.13, nearly half its average value just one quarter before the stress test period. Post-crisis, Basel III reforms have required banks to improve their capital positions.[13] One such reform, the capital conservation buffer, has been enforceable since 2019 and aims to add an extra layer of resilience to the banking system in the face of unexpected and significant losses. The minimum capital buffer was set at 2.5% of total risk assets on top of the common equity tier 1 capital (CET1), raising the minimum T1CR for large BHCs subject to a stress test to 7%. Since BHCs subject to stress tests are large banks, many also meet the 7% minimum T1CR requirement for both methods throughout the stress test horizon. The aggregate T1CR reaches a minimum of 7.70% for the Best ML and 8.22% for the FE-LM model, comfortably above the 7% treshold. However, the impact of severely adverse conditions is not uniform across all banks for both models. For instance, Northern Trust Corporation is less affected by the simulated crisis scenarios than Bank

[13] These reforms raised the minimum requirement for different regulatory capital and introduced other capital buffers, improving the capital position of banks.

Table 15
Out-of-sample relative mean square error of PPNR's forecast models under specification (1) across various forecast horizons and different forecasting techniques (only banks selected for the empirical stress test).

Model	h = 1	h = 2	h = 3	h = 4	h = 5	h = 6	h = 7	h = 8	h = 9
Benchmark model	*1*	*1*	*1*	*1*	*1*	*1*	*1*	*1*	*1*
Pooled OLS	0.37	0.39	1.02	0.40	0.37	0.38	0.39	0.47	0.45
Autoregressive model	3.79	2.61	2.44	3.20	1.95	1.59	0.61	1.94	1.55
Random walk	4.05	2.98	2.79	4.00	2.47	2.10	2.21	2.80	2.16
Lasso	0.32	0.34	0.33	0.37	0.33	0.34	0.35	0.40	0.39
Adaptive Lasso	**0.26*****	**0.26*****	**0.29*****	**0.31*****	0.29	0.31	**0.31****	**0.32*****	**0.31*****
Ridge	0.74	0.55	0.50	0.60	0.48	0.45	0.45	0.53	0.51
Elastic net	0.37	0.36	0.37	0.37	0.39	0.38	0.40	0.47	0.46
Principal component	4.16	2.98	2.75	3.46	2.21	1.87	1.97	2.35	1.88
NN1	0.89	0.61	0.60	0.51	0.47	0.73	0.63	0.61	0.70
NN2	1.05	0.41	0.65	0.51	0.50	0.45	0.50	0.70	0.63
NN3	1.74	1.70	0.70	0.70	0.55	0.70	0.71	0.80	0.50
Gradient boosting	0.53	0.27	0.27	0.43	0.35	0.54	0.36	0.46	0.35
Random Forest	0.31	0.29	0.31	0.40	**0.19*****	**0.29*****	0.35	0.40	0.34

Note: (1) Machine learning models and standard linear models include the same variables that FED uses in the bank's stress test. (2) The first line represents the mean square error (MSE) of the benchmark model normalized to 1, while the other lines present the relative MSE of different forecast models. (3) The figures in bold relate to the best forecast model. No star : best linear models outperform the best machine learning model or have the same performance as it for DM test, *: best machine learning model outperforms best standard linear model for the DM test at 10% level , **: 5% level ; ***: 1% level. NN1 designates the neural network with one hidden layer that has 32 neurons. NN2 is the neural network with two hidden layers containing 32 and 16 neurons, respectively. NN3 is a neural network with three hidden layers having 32, 16, and 8 neurons, respectively.

Table 16
Out-of-sample relative mean square error of PPNR's forecast models under specification (2) across various forecast horizons and different forecasting techniques (only banks selected for the empirical stress test).

Model	h = 1	h = 2	h = 3	h = 4	h = 5	h = 6	h = 7	h = 8	h = 9
Benchmark model	*1*	*1*	*1*	*1*	*1*	*1*	*1*	*1*	*1*
Pooled OLS	0.37	0.39	1.02	0.40	0.37	0.38	0.39	0.47	0.45
Autoregressive model	3.79	2.61	2.44	3.20	1.95	1.59	0.61	1.94	1.55
Random walk	4.05	2.98	2.79	4.00	2.47	2.10	2.21	2.80	2.16
Lasso	0.39	0.30	0.63	0.37	0.29	0.30	0.32	0.36	0.34
Adaptive Lasso	**0.31****	**0.26*****	1.77	**0.28*****	**0.24*****	0.27	0.57	**0.34*****	0.44
Ridge	0.77	0.55	0.50	0.58	0.45	0.45	0.47	0.50	0.65
Elastic net	0.41	0.43	1.23	0.38	0.42	0.43	0.48	0.51	0.51
Principal component	1.16	0.50	0.81	1.00	0.95	0.61	1.97	0.39	0.60
NN1	0.92	0.61	0.80	0.47	0.40	0.63	0.65	0.57	0.55
NN2	1.20	0.41	0.75	0.45	0.43	0.35	0.55	0.70	0.60
NN3	1.85	1.70	0.90	0.60	0.50	0.60	0.72	0.74	0.53
Gradient boosting	0.46	0.33	**0.29*****	0.36	0.42	0.33	0.35	0.36	0.32
Random Forest	0.36	**0.26*****	0.31	0.43	0.36	**0.23*****	**0.28*****	0.35	**0.28*****

Note: (1) Machine learning models are estimated with extensive macroeconomic database.(2) The first line represents the mean square error (MSE) of the benchmark model normalized to 1, while the other lines present the relative MSE of different forecast models. (3) The figures in bold relate to the best forecast model. (4) No star : Best linear models outperform the best machine learning model or have the same performance as it for the DM test *: Best machine learning model outperforms best standard linear model for the DM test at 10% level ; **: 5% level ; ***: 1% level. NN1 designates the neural network with one hidden layer that has 32 neurons. NN2 is the neural network with two hidden layers containing 32 and 16 neurons, respectively. NN3 is a neural network with three hidden layers having 32, 16, and 8 neurons, respectively.

of America Corporation for the Best ML model. Conversely, JP Morgan Chase and Company experiences a more pronounced impact from the simulated adverse economic conditions than Well Fargo in the FE-LM model. Business model differences could explain the crisis's disparate effects on these banks.

5.2.2. The analysis of banks' vulnerabilities : Best ML versus FE-LM modeling

In the following analysis, I will delve deeper into the T1CR distribution by also focusing on the range of possibilities that it represents for each bank. Table 21 details the probability of each bank violating specific T1CR thresholds in 2021q1, according to both models. The "All" row represents the probability of the aggregate T1CR falling below-given thresholds for both models, effectively capturing the average risk of violation for all banks. As the results indicate, there is, on average, no risk of the banks not meeting the 4.5% minimum T1CR threshold according to both models. However, this average needs to capture the nuances across different banks and models. For example, the Best ML model indicates a 9% chance of Bank of America violating the 4.5% minimum requirement, while it shows a 0% chance of such a violation for State Street Corporation. Similarly, Citigroup INC has a 7% risk of not meeting the rule according to the Best ML model, but this risk drops to zero with the FE-LM model. When considering the 7% minimum requirement, the picture changes somewhat. Only a few banks are predicted to have no chance of violating this threshold. Nevertheless, the overall probability of violating the 7% requirement is estimated at 2% for both models. This result aligns with the findings of Covas et al. (2014), who estimated a 2.5% chance of breaking an 8% T1CR minimum requirement for both the fixed effect quantile and fixed effect linear models. This analysis suggests that at the 4.5% threshold, most banks have a negligible risk of failing to meet the minimum T1CR rule according to both models. However, at 7%, there is a generally higher risk of banks not meeting the requirement.

Table 22 provides a more in-depth look at the two models' conditional distribution of T1CR at the end of the stress test horizon. The probability density function (PDF) of T1CR is left-skewed, indicating an asymmetric distribution. This pattern is consistent with some studies (such as Covas et al. (2014) and Hirtle et al. (2016)) that find a left-skewed T1CR PDF under crisis scenarios. A closer examination reveals

Table 17
Out-of-sample relative mean square error of PPNR's forecast models under specification (3) across various forecast horizons and different forecasting techniques (only banks selected for the empirical stress test).

Model	h = 1	h = 2	h = 3	h = 4	h = 5	h = 6	h = 7	h = 8	h = 9
Benchmark model	*1*	*1*	*1*	*1*	*1*	*1*	*1*	*1*	*1*
Pooled OLS	0.37	0.39	1.02	0.40	0.37	0.38	0.39	0.47	0.45
Autoregressive model	3.79	2.61	2.44	3.20	1.95	1.59	0.61	1.94	1.55
Random walk	4.05	2.98	2.79	4.00	2.47	2.10	2.21	2.80	2.16
Lasso	0.35	0.35	0.69	0.37	0.36	**0.30*****	0.46	**0.39*****	**0.36*****
Adaptive Lasso	**0.30*****	**0.30*****	1.75	**0.29*****	**0.25*****	**0.30*****	0.77	0.41	0.50
Ridge	0.75	0.59	0.50	0.55	0.45	0.48	0.57	0.55	0.70
Elastic net	0.39	0.43	1.19	0.37	0.29	0.39	0.48	0.50	**0.36*****
Principal component	1.16	0.91	0.87	1.09	0.72	0.70	0.64	0.76	0.60
NN1	0.90	0.70	0.81	0.50	0.50	0.65	0.70	0.70	0.55
NN2	1.10	0.55	0.75	0.60	0.40	0.37	0.59	0.75	0.60
NN3	1.80	1.60	0.90	0.65	0.60	0.62	0.72	0.79	0.53
Gradient boosting	0.84	0.37	**0.31*****	0.43	0.32	0.32	**0.36**	0.43	0.37
Random Forest	0.47	0.39	0.49	0.77	0.47	0.43	0.52	0.70	0.55

Note: (1) Machine learning models include extensive macroeconomic database and enriched BHC characteristics .(2) The first line represents the mean square error (MSE) of the benchmark model normalized to 1, while the other lines present the relative MSE of different forecast models.(3) The figures in bold relate to the best forecast model. (4) no star : Best linear models outperform the best machine learning model or have the same performance as it for the DM test *: Best machine learning model outperforms best standard linear model for the DM test at 10% level; **: significance at 5% ; ***: significance at 1%. NN1 designates the neural network with one hidden layer that has 32 neurons. NN2 is the neural network with two hidden layers containing 32 and 16 neurons, respectively. NN3 is a neural network with three hidden layers having 32, 16, and 8 neurons, respectively.

Table 18
Percentage variation of best ML relative to best SL model forecast accuracy of PPNR under the three specifications (only banks selected for the empirical stress test).

Measure	h = 1	h = 2	h = 3	h = 4	h = 5	h = 6	h = 7	h = 8	h = 9
$PC_1(h)$	−29.73	−33.33	−71.57	−22.50	−48.65	−23.68	−20.51	−31.91	−31.11
$PC_2(h)$	−16.21	−33.33	−71.57	−30.00	−35.14	−39.47	−28.20	−27.66	−37.78
$PC_3(h)$	−18.92	−23.08	−69.61	−27.50	−32.43	−21.05	−7.69	−17.02	−20.00

Note: $PC_j(h)$, j = 1,2,3 denotes the percent variation of best ML relative to best SL forecast accuracy for different forecast horizons h and for specification j. A negative value means that using the best ML reduces the RMSE of the best SL of PC_j, improving its forecast accuracy. A positive value on contrary denotes a deterioration of forecast accuracy in case of using best ML model instead of best SL model.

Table 19
Severely adverse scenarios in 2019q1–2021q1.

Date	GDP	UNRATE	Inflation rate	TB3MS	GS5	GS10	Mortgage rate	SP500	USSTHPI	VXOCLS
2018q4	1.31	3.83	1.29	2.32	2.88	3.03	4.78	−22.95	−0.18	43.1
2019q1	−5.0	4.7	1.2	0.3	0.3	0.8	3.9	−36.6	−3.0	44.4
2019q2	−9.4	6.3	1.6	0.2	0.5	0.9	4.2	−19.4	−3.1	43.1
2019q3	−7.2	7.5	1.7	0.1	0.6	1.0	4.4	−9.8	−3.7	39.2
2019q4	−5.0	8.4	1.8	0.1	0.6	1.1	4.5	−3.5	−4.4	34.9
2020q1	−3.8	9.2	1.9	0.1	0.7	1.2	4.3	4.6	−4.6	30.5
2020q2	−1.5	9.7	1.8	0.1	0.7	1.2	4.2	6.9	−4.2	27.3
2020q3	−0.3	10.0	2.0	0.1	0.7	1.2	4.1	7.1	−4.4	25.3
2020q4	2.9	9.9	2.0	0.1	0.7	1.2	3.9	8.5	−2.6	23.5
2021q1	3.6	9.7	2.1	0.1	0.9	1.5	3.9	5.4	−0.7	22.5

Table 20
Predicted T1CR and historical values.

Bank	min T1CR on the stress horizon		$T1CR_{2018q4}$	min T1CR for G.R	$T1CR_{2007q3}$
	Best ML	FE-LM			
Bank of America Corporation	7.44	8.28	13.66	6.17	6.63
Bancwest Corporation	6.53	4.50	11.01	3.95	4.62
BBVA USA Bancshares INC	5.90	7.83	12.05	7.81	8.56
Citigroup INC	7.36	10.44	14.83	6.10	7.13
Comerica INC	7.85	6.61	10.25	6.70	6.96
Fifth Third Bancorp	7.77	8.97	11.25	5.82	6.57
JP Morgan Chase and Company	7.61	6.01	13.67	7.25	7.42
Keycorp	6.05	5.05	10.56	5.94	6.43
Huntington Bancshares INC	8.70	4.55	10.63	6.22	7.24
Northern Trust Corporation	13.48	10.44	14.50	8.57	9.46
PNC Financial Services INC	7.61	8.37	12.01	4.47	6.28
State Street Corporation	17.30	12.39	17.55	9.06	11.61
Well Fargo	7.62	9.01	13.48	6.29	7.68
All	7.70	8.22	13.51	6.64	7.13

Table 21
Probability of violating different thresholds in 2021q1.

Bank	Minimum T1CR requirement: 4.5%		Minimum T1CR requirement: 7%	
	Best ML	FE-LM	Best ML	FE-LM
Bank of America Corporation	0.09	0.00	0.45	0.01
Bancwest Corporation	0.00	0.51	0.23	0.93
BBVA USA Bancshares INC	0.01	0.07	0.46	0.39
Citigroup INC	0.07	0.00	0.44	0.00
Comerica INC	0.06	0.00	0.39	0.38
Fifth Third Bancorp	0.07	0.00	0.40	0.00
JP Morgan Chase and Company	0.04	0.00	0.32	0.38
Keycorp	0.00	0.25	0.44	0.92
Huntington Bancshares INC	0.00	0.57	0.01	0.97
Northern Trust Corporation	0.00	0.00	0.00	0.00
PNC Financial Services INC	0.08	0.00	0.47	0.09
State Street Corporation	0.00	0.00	0.00	0.00
Well Fargo	0.08	0.00	0.39	0.00
All	0.00	0.00	0.02	0.02

Note: The probability to violate the given threshold is the probability of having T1CR strictly below that value.

Table 22
Predicted conditional percentiles of T1CR in the stress test horizon in 2021q1.

Bank	1st percentile		5th percentile		Median		95th percentile	
	Best ML	FE-LM	Best ML	FE-LM	Best ML	FE-LM	Best ML	FE-LM
Bank of America Corporation	−0.34	6.73	7.40	7.63	8.55	8.95	10.02	10.36
Bancwest Corporation	6.24	1.05	6.72	2.87	7.43	4.58	8.47	7.31
BBVA USA Bancshares INC	4.43	−0.34	6.09	7.05	6.79	8.38	9.16	9.99
Citigroup INC	−0.40	9.30	7.40	9.79	8.47	10.98	9.89	12.27
Comerica INC	−0.70	4.95	7.35	6.09	8.32	7.04	9.80	8.23
Fifth Third Bancorp	−0.34	7.05	7.38	7.85	8.39	9.04	9.99	10.36
JP Morgan Chase and Company	−0.50	4.95	7.46	6.16	8.58	7.17	10.36	9.06
Keycorp	4.47	2.18	6.03	3.80	6.79	6.10	9.39	7.61
Huntington Bancshares INC	6.75	1.49	7.63	3.02	8.79	4.52	10.33	6.84
Northern Trust Corporation	11.10	9.30	12.03	9.79	13.60	10.98	15.34	12.26
PNC Financial Services INC	−0.40	4.41	7.39	7.42	8.39	8.68	9.96	10.33
State Street Corporation	12.43	9.06	14.02	11.59	17.44	12.96	20.02	15.53
Well Fargo	−0.60	7.54	7.46	8.08	8.56	9.29	10.30	10.64
All	6.74	6.74	7.29	7.40	8.87	9.42	12.25	12.40

that for almost all global systemically important banks (G-SIBs),[14] the left tail of the T1CR distribution is more substantial for the Best ML model than the FE-LM model. This fact is visually evident in Fig. 3, which plots the probability densities of T1CR for the four G-SIBs. Covas et al. (2014) found a similar phenomenon, where the left tail of the T1CR at the end of their stress-test period (2013q4) was denser for their nonlinear baseline model than for the fixed effect linear model for most banks. One plausible explanation for this could be a heavier right tail in the conditional distribution of NCO under the Best ML model compared to the FE-LM model, as shown in Fig. 4. When looking at PPNR, another component of T1CR, Fig. 5, does not depict a distinct difference between the left tail distributions for the two models. These findings suggest that the linear framework may underestimate the risk of significant losses for G-SIBs during economic downturns compared to a framework combining nonlinear models for loss predictions and linear models for revenue predictions. Consequently, linear models might paint a more optimistic picture of banking stability and minimize the perceived systemic risk during economic downturns.

[14] A global systemically important bank is a financial institution whose failure increases systemic risk and may trigger a financial crisis. The following banks in our sample figure out in the list for the year 2019 of the globally systemically banks established annually by the Financial Stability Board in coordination with the Basel committee: Bank of America Corporation, Citigroup INC, JP Morgan Chase and Company, State Street Corporation, and Well Fargo. Bank of America Corporation, Citigroup INC, JP Morgan Chase and Company, and Well Fargo are the larger banks in our sample, weighing 80% of the sampled bank's total assets.

5.2.3. Predicting capital shortfalls under distressed macroeconomic and financial conditions

To ascertain the capital gap for a given threshold, I quantify the discrepancy between the average predicted capital and the indispensable minimum capital required to prevent the breach of the specified criteria. I establish the aggregate capital shortfall by compiling the capital gaps across all banks. This calculation is illustrated in Table 23. At the end of the stress assessment period, with a minimum capital requirement set at 4.5%, there is no aggregate accumulated capital gap. However, the aggregate capital shortfall reached 1 billion USD during the Great Recession. Raising the threshold to 7% significantly inflates the aggregate cumulative capital shortfall to 2 billion USD under the Best ML model and a staggering 28 billion USD under the FE-LM model. In stark contrast, the capital gap surged to 85 billion USD under the same 7% threshold during the Great Recession. These results underscore the enhanced stability of the financial system post-Great Recession. This improvement is attributable, in part, to more rigorous regulations and increased frequency of financial oversight exercises. Thus, this analysis offers a robust argument for the efficacy of regulatory reform in mitigating financial system vulnerabilities.

5.2.4. Comparing predicted T1CR PDF accuracy between best ML and FE-LM models during the great recession

I use two evaluation methods to compare the two models: interval forecast evaluation and direct density forecast evaluation. In particular, I concentrate on one quarter-ahead forecast.

Interval forecast evaluation

Evaluating interval forecasts to rank predicted densities is a common approach in the literature (see Gneiting and Raftery (2007) and Goulet Coulombe et al. (2022)). I use three criteria: empirical coverage rate, average interval length, and average interval score.

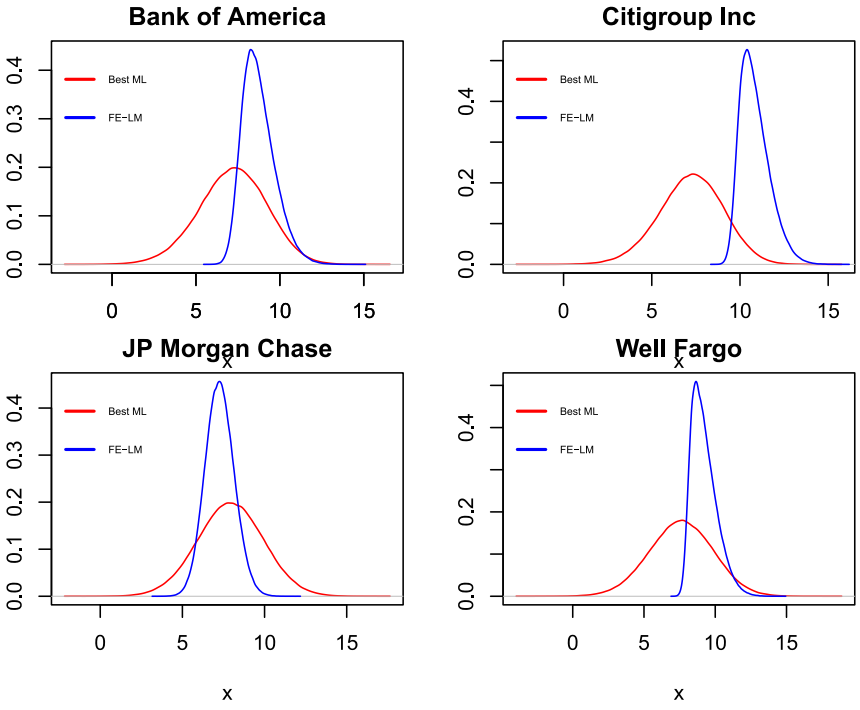


Fig. 3. Probability Densities of T1CR for the largest G-SIBs bank in 2021q1. Note : These figures show the estimated conditional skewed t PDF at the end of the stress test period for the four largest banks of the sample.

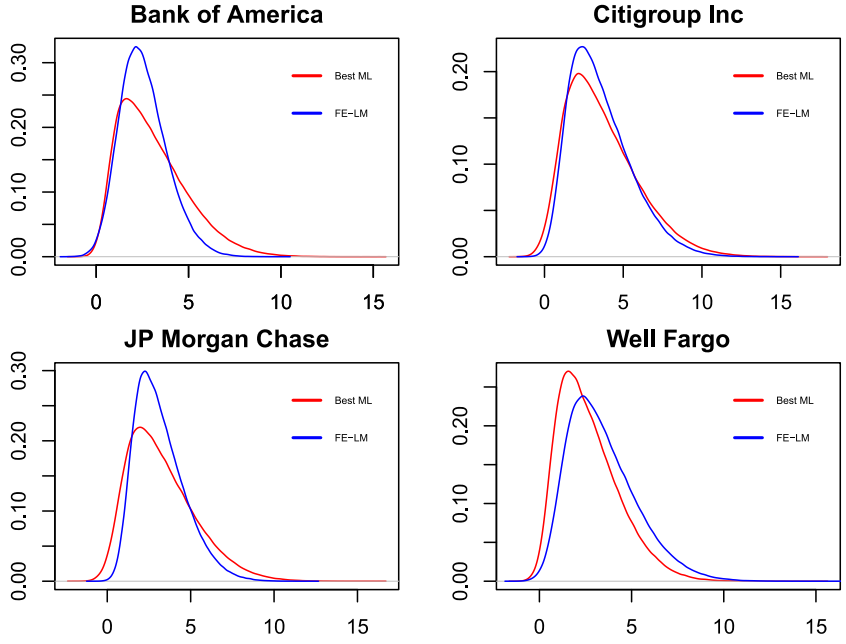


Fig. 4. Probability Densities of NCO for the largest G-SIBs banks in 2021q11.

Empirical Coverage Rate: This measures how often the actual value of T1CR falls within the predicted interval, calculated as $(1-\alpha)100\%$, where α signifies the significance level. The predictive quantiles at level $\frac{\alpha}{2}$ and $1-\frac{\alpha}{2}$ denote the lower and upper limits of the interval, labeled as l and u, respectively. I set α at 0.1 for my evaluation. A higher (or lower) coverage rate than 90% implies that the predicted interval is overly wide (or narrow). The most accurate model maintains an empirical coverage rate closest to 90%.

Average Interval Length: This is the difference between the upper and lower limits of the interval. I ascertain the average length across all forecasted intervals.

Average Interval Score: Introduced by Gneiting and Raftery (2007), this criterion gauges the quality of interval forecasts, tracing its roots back to Winkler and Murphy (1979). The interval score, denoted as $S_{\alpha}^{int}(l,u,x)$, where x is a realized value of T1CR, is calculated as follows:

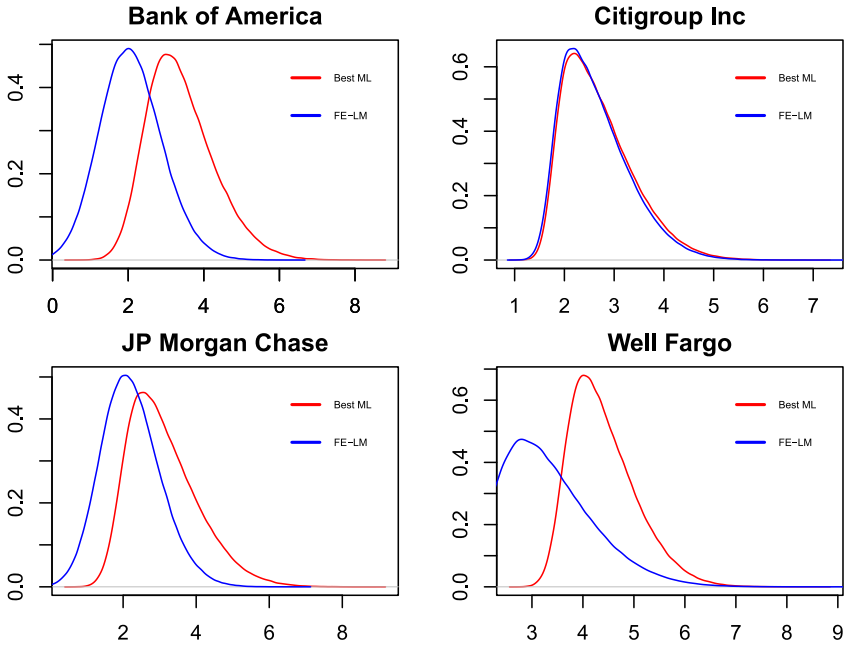


Fig. 5. Probability Densities of PPNR for the largest G-SIBs banks in 2021q1.

Table 23
The cumulated capital gap at the end of the Great Recession[c] (2009q2) and the stress test period (2021q1) (billion of USD dollars).

Bank	Stress test horizon				Great Recession	
	4.5% T1CR requirement		7% T1CR requirement		4.5% T1CR requirement	7% T1CR requirement
	Best ML	FE-LM	Best ML	FE-LM		
Bank of America Corporation	0.00	0.00	0.00	0.00	0.00	22.30
Bancwest Corporation	0.00	0.00	0.06	0.90	1.02	11.50
BBVA USA Bancshares INC	0.00	0.00	0.95	0.00	0.00	0.00
Citigroup INC	0.00	0.00	0.00	0.00	0.00	17.90
Comerica INC	0.00	0.00	0.00	0.26	0.00	0.65
Fifth Third Bancorp	0.00	0.00	0.00	0.00	0.00	4.46
JP Morgan Chase and Company	0.00	0.00	0.00	15.20	0.00	0.00
Keycorp	0.00	0.00	1.52	4.66	0.00	3.29
Huntington Bancshares INC	0.00	0.00	0.00	7.40	0.00	0.66
Northern Trust Corporation	0.00	0.00	0.00	0.00	0.00	0.00
PNC Financial Services INC	0.00	0.00	0.00	0.00	0.03	9.63
State Street Corporation	0.00	0.00	0.00	0.00	0.00	0.00
Well Fargo	0.00	0.00	0.00	0.00	0.00	15.30
All	0.00	0.00	2.53	28.42	1.05	85.69

Note: During the Great Recession, T1CR requirements were not set. We use these thresholds to evaluate the capital gap prevailing at that period if the conditions were applicable.

$$S_{\alpha}^{int}(l,u,x) = (u-l) + \frac{2}{\alpha}(l-x)1x < l + \frac{2}{\alpha}(x-u)1x > u. \quad (9)$$

Here, 1(.) represents the indicator function. According to this metric, the superior model yields the smallest score value. Intuitively, the interval score encourages a narrower interval forecast, rewarding precision and accuracy. This intuition can be traced to Engle (1982), who advocated for dynamic interval forecasts surrounding point predictions. He argued that these intervals should contract during stable periods and expand during turbulent times due to the inherent stability leading to more accurate forecasts. The score also penalizes forecasters for missing the interval, with the severity of the penalty varying with α.

When comparing models, the Best ML approach more accurately approximates the T1CR density during the 2007–2009 financial crisis in terms of empirical coverage rate and average interval score, as illustrated in Table 24. However, it is worth noting that the FE-LM model exhibits a shorter average interval length than the Best ML model. This observation implies that, on average, the FE-LM model predicts a more narrowly distributed left tail of the T1CR density during the Great Recession compared to the Best ML model. Thus, while both models have their strengths, the Best ML model's superior performance in both empirical coverage rate and average interval score suggests it is generally more effective in modeling and predicting T1CR density during financial crises. However, the thinner left tail produced by the FE-LM model can be advantageous under certain circumstances and should be noticed.

Table 24
Comparison of one step ahead 90% interval forecasts between ML and linear modeling.

Model	Empirical coverage rate (%)	Average interval length	Average interval score
Best ML modeling	80.95	3.34	7.59
FE-LM modeling	50.00	2.20	9.37

Direct density forecast evaluation

Guided by Goulet Coulombe et al. (2022) methodology, I employ a scoring method to gauge the accuracy of density forecasts. The score is

Table 25
Score for selected stressed BHC throughout the 2007–2009 financial crisis.

Bank holding companies	Score for ML modeling (S_{ML})	Score for linear modeling (S_L)	Difference (S_{ML}-S_L)
Bank of America Corporation	2.17	77.84	−75.67
Bancwest corporation	1.47	8.21	−6.74
BBVA USA Bancshares Inc	1.18	39.94	−38.76
Citigroup INC	3.40	52.15	−48.75
Comerica Inc	3.37	127.73	−124.36
Fifth Third Bancorp	1.96	38.00	−36.04
JP Morgan chase and company	1.58	29.99	−28.41
Keycorp	2.20	8.23	−6.03
Huntington Bancshares Inc	1.99	7.19	−5.20
Northern trust corporation	1.86	2.19	−0.33
PNC Financial Services INC	1.50	27.25	−25.76
State street corporation	2.47	2.65	−0.18
Wells Fargo et Company	1.23	26.18	−24.95
All	2.03	37.04	**−35.01*****

Note: *** indicates that the ML modeling is better than the Linear modeling in predicting T1CR densities at 1% significant level.

given as:

$$S_{i,t+1} = -log(\hat{f}(y_{i,t+1}/X_{i,t})). \tag{10}$$

In this equation, $\hat{f}(y_{t+1})$ represents the predicted conditional density, evaluated at the realized value $y_{i,t+1}$. The indices i and t are the cross-sectional and time dimensions of the panel, respectively, while $X_{i,t}$ is the set of information available at quarter t. The model with the lowest average score is considered the most accurate, as a lower score signals a higher mass closer to the realized value. Table 24 reveals the average score for each selected bank during the 2007–2009 financial crisis, showing the Best ML modeling with a lower average score than the FE-LM modeling. To assess the comparative ranks of the two models, I employ the Diebold–Mariano test (DM test), modifying the definition of the loss differential as follows:

$$d_{i,t} = S_{i,t}^{ML} - S_{i,t}^{LIN}; \tag{11}$$

where $S_{i,t}^{ML}$ and $S_{i,t}^{LIN}$ denote the scores for the Best ML and FE-LM modeling for bank i at quarter t, respectively. Following Iacopini, Ravazzolo, and Rossini (2023) interpretation, $d_{i,t}$ serves the same function as in the original DM test. I test the null hypothesis of equal forecast accuracy $H0 : E(di,t) = 0$ against the alternative hypothesis $H_1 : E(d_{i,t}) < 0$, favoring the Best ML model. I perform this test by regressing $d_{i,t}$ on a constant, where a zero constant corresponds to H_0, and a negative constant signifies the superiority of the Best ML model over FE-LM modeling for predicting T1CR density during the Great Recession. Table 25 displays a significantly negative constant at the 1% significance level, suggesting that the Best ML model is superior to the FE-LM model in approximating T1CR density during the 2007–2009 financial crisis.

6. Discussions on the implications of the results

This section highlights the policy implications of the results for the regulatory authorities and the banks. It also raises some issues on the applicability and transparency of the machine learning modeling.

6.1. For regulatory authorities

Results show that the machine learning models better predict two key banking variables in the stress tests, PPNR and NCO, compared to standard linear models. From this, regulators can use these models to predict bank performance more accurately under adverse economic conditions. These findings align with the findings of Khandani, Kim, and Lo (2010), who demonstrate the effectiveness of machine learning in financial risk prediction. Furthermore, by showing that ML models better capture risk in the stress test than linear models, this paper provides avenues for regulatory authorities to use these advanced tools in risk evaluation. Accordingly, regulators can identify potential risks and vulnerabilities in the banking sector, leading to more informed regulatory decisions. James, Witten, Hastie, Tibshirani, et al. (2013) has emphasized the capacity of ML models to handle complex nonlinear relationships. Moreover, Machine learning models can be tailored to analyze specific aspects of bank operations, allowing for more targeted and effective regulatory interventions. This customization ensures that regulatory actions are broad-based and intensely focused where needed. The idea of tailoring machine learning models for specific analytical needs reflects the views of Murphy (2012), emphasizing the versatility of these models in diverse applications. The results also point to the importance of big data in predicting PPNR and NCO for some forecast horizons. These findings suggest that regulatory authorities explore additional data to improve the forecast accuracy of banking variables and, therefore, stress test outputs. Varian (2014) highlighted the tremendous predictive power of ML combined with big data. The other exciting feature of ML is its flexibility and adaptability. Machine learning algorithms can adapt and learn from new data, making stress testing a dynamic and continuously evolving process. This adaptability central to Bishop and Nasrabadi (2006) ensures that stress tests remain relevant and effective in changing economic conditions and emerging financial trends.

6.2. For banks

Banks can use machine learning models better to understand their risk exposure under various stress scenarios. This improved insight allows for more effective risk management strategies, helping banks to better prepare for potential adverse economic conditions. Jorion (2007) sheds light on the importance of accurate risk assessment in banking. Furthermore, with more accurate forecasts of PPNR and NCO, banks can better estimate their capital requirements. This exercise helps ensure they have sufficient capital buffers to withstand economic downturns, aligning with regulatory expectations and reducing the risk of insolvency. This proactive approach can position them favorably with regulators and lead to a more constructive and collaborative regulatory relationship. Similarly, Hull (2012) underlines the importance of accurate capital planning for banks. Machine learning models can also provide banks with deeper insights into which business lines or products are most vulnerable in times of stress. This information can guide strategic decision-making, such as where to allocate resources or which areas to diversify. The decision-making improvements through machine learning can be tied back to the principles outlined by Simon (1997), highlighting how data-driven approaches refine strategic choices. In addition, Machine learning enables more dynamic modeling of asset and liability portfolios under various scenarios, allowing banks to adjust their portfolios proactively in response to changing economic conditions. De Prado (2018) underline this dynamic portfolio management through ML.

6.3. Some challenges of ML approach

ML's approach offers real benefits in the stress test for regulatory authorities and banks. However, the use of ML modeling poses three main challenges. First, Molnar (2020) highlights the issue of interpretability, particularly in complex models like deep learning. To overcome this issue, Burrell (2016) calls for transparency in machine learning applications in finance. This transparency reinforces the need for clear, data-driven insights in regulatory practices, ensuring stakeholder trust in the system. Second, Dhar (2013) notes that the effectiveness of machine learning models heavily depends on the quality and quantity of available data. This requirement can be a significant limitation in the banking sector, especially with sensitive or proprietary financial data. Nevertheless, the increased availability and transparency of financial information overcomes this challenge—for example, the FED Chicago quarterly releases granular data on US banks' balance sheets. The FRED database releases hundreds of monthly and quarterly financial and macroeconomic series. Last, another challenge of using ML is computational cost and efficiency. Jordan and Mitchell (2015) underlines the computational costs associated with training sophisticated models, which can be a barrier for some banks. However, the increased sophistication of computers has significantly reduced this cost. This cost reduction gives sophisticated advanced ML tools an advantage over a more straightforward linear model.

7. Conclusion

This paper thoroughly explores the benefits of integrating machine learning (ML) methodologies with comprehensive macroeconomic and bank-specific data, intending to optimize risk analysis during stress tests. The research unfolds in two sequential stages, yielding substantive findings contributing to our risk analysis understanding.

In the first stage, I establish the superior efficacy of Random Forest, a nonparametric and nonlinear ML technique, over Pooled OLS and other ML models in predicting Net Charge-Offs (NCO) across nine forecast horizons. This finding resonates with Covas et al. (2014)'s emphasis on the nonlinear relationship between bank losses and predictors and further elaborates on the utility of Random Forest in enhancing the Relative Mean Squared Error (RMSE) of linear models in NCO forecasting for stress tests. Nonetheless, the predictive power of NCO forecasting was further improved, by approximately 17%, across six of nine forecast horizons when I expanded the macroeconomic database. Conversely, supplementing bank balance sheet information has no beneficial impact and, in some cases, impedes the predictive capabilities of Random Forest. Regarding Pre-Provision Net Revenue (PPNR), my findings lean towards Adaptive Lasso, a linear ML model that outperforms standard linear models across all forecast horizons. The linear relationship between PPNR and predictors persists, yet the regularization imparted by Adaptive Lasso triggers a notable improvement over linear models. While using a larger macroeconomic database enhances the best ML models' performance on horizons 5, 6, and 7, additional bank balance sheet information had a detrimental effect on the performance of ML models.

A critical revelation of the paper lies in analyzing the Tier 1 Capital Ratio (T1CR) distribution in the second stage. The left tail of the T1CR distribution is considerably heavier when ML models are employed compared to linear models, especially for globally systemically important banks. Moreover, ML models offer a superior approximation of T1CR density, particularly under strained macro-financial conditions. These findings underline the risk of over-optimism and systemic risk underestimation when exclusively relying on linear models. Thus, ML models, with their ability to incorporate nonlinearity and complexity, provide a more holistic perspective on the interconnectedness of the banking sector and potential systemic risks.

This paper underpins the benefits of ML models in enhancing stress test risk analysis and asserts the significance of an extensive macroeconomic database for NCO and PPNR forecasts. Nonetheless, the study does not delve into the direct influence of additional predictors on the accuracy of T1CR density forecast, paving the way for future research to pinpoint additional predictors for NCO and PPNR models and assess their implications for T1CR forecasts. This avenue of investigation can deepen our understanding of the direct effects of these predictors on the efficacy of stress test methodologies.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work, I used chatgpt in order to improve language and readability. After using this tool, I reviewed and edited the content as needed and takes full responsibility for the content of the publication.

Appendix A. Supplementary data

This appendix provides details on the data source, computation of variables, and the method of treatment of outliers.

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.irfa.2024.103476.

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